Arab American University

Faculty of Engineering and Information Technology

Computer Systems Engineering

SENIOR PROJECT I

KidSentinel

(*Smart Child Monitoring System*)

2026

Students Statement

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Computer Systems Engineering Dept.
Submitted in partial fulfillment of the requirements of B.Sc. Degree in Computer Systems Engineering

June, 2026

Supervisor Certification

This to certify that the work presented in this senior year project manuscript was carried out under my supervision, which is entitled:

"KidSentinel"

I hereby that the aforementioned students have successfully finished their senior year project and by submitting this report they have fulfilled in partial the requirements of B.Sc. Degree in Computer systems Engineering.

I also, hereby that I have **read, reviewed and corrected the technical content** of this report and I believe that it is adequate in scope, quality and content and it is in alignment with the ABET requirements and the department guidelines.

ACKNOWLEDGMENT

As we are finishing our project. We want to say thank you to everyone who helped us in ways while we were doing this research.

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Secondly, we want to thank our project advisor, Prof. Hazem Khanfar for helping us and motivating us while we were working on this project.

Finally, we want to thank our family members, our friends and our colleagues for motivating the project team throughout the thing.

ABSTRACT

With the increasing number of children in nurseries, monitoring has turned into a difficult process, as it requires constant attention by the supervisor and takes a lot of time, particularly in cases where several children require monitoring simultaneously. This problem might be even worse in the situation where the children are sleepy, have abnormal movement, or are potentially exposed to danger. Therefore, the necessity has emerged to develop an intelligent system to make the monitoring more effective and increase the safety of children.

The current project suggests developing an intelligent system that will allow monitoring the child in the stroller continuously, providing parents and administrative personnel with information on the condition of the child using a special mobile application. The system will utilize special sensors to ensure that the child remains in the stroller and to define whether he or she is sleeping or awake. In addition, the system will identify the particular child using their individual information and link it with the information collected about the child.

Finally, the system will feature video observation via the camera that will be installed in the stroller. It will immediately send a notification if something changes, e.g., the child starts sleeping, encounters a threat or loses the signal. Moreover, as a part of the security mechanism, the stroller will automatically stop its motion in case the child faces potential threats.

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LIST OF SYMBOLS AND ABBREVIATIONS

LIST OF SYMBOLS	
V	Volt
mA	Milliampere
A	Ampere
cm	Centimeter
LIST OF ABBREVIATIONS	
IoT	Internet of Things
NFC	Near Field Communication
FSR	Force Sensing Resistor
MPU	Motion Processing Unit
ESP	Espressif Systems Platform
CAM	Camera Module
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation
SDA	Serial Data Line
SCL	Serial Clock Line
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver-Transmitter
ADC	Analog to Digital Converter
RFID	Radio Frequency Identification
JSON	JavaScript Object Notation
AC	Alternating Current
DC	Direct Current
I2C	Inter-Integrated Circuit

1 CHAPTER 1: INTRODUCTION

Between 1990 and 2010 the United States saw a lot of child injuries from stroller accidents. There were more than 360,000 cases [1]. These accidents happened when the stroller tipped over or fell. Kids got hurt on their heads or faces.

These numbers are scary. They show that strollers can be bad for kids if they are not watched closely. So, it is very important to make sure kids are safe in nurseries. We are working on a project to make a system that keeps an eye on kids in the nursery all the time. This system checks on the kids. Send messages to their parents and the nursery staff if something is wrong.

The stroller accidents show that we need to be careful. The system we are making wants to make sure kids are safe and sound in the nursery. If we use strollers, the way they can be safe. Having a smart system to watch the kids can help prevent accidents and make everyone feel better about the kids being safe. The smart system is an idea because it helps reduce the risks of stroller accidents and makes the nursery a safer place for kids.

1.1 Problem Statement and Purpose

In a nursery, the kids may not get the attention they need all the time, especially when they are near something that can hurt them or when they bump into another kid. This can cause injuries that the teachers may not see away.

When the kids are sleeping the teachers may not be able to watch them all the time so they might not notice if something is wrong with them.

The parents do not have a way to check on their kids during the day; they must wait for the nursery staff to tell them what is going on. This can take a long time, and the parents may not get the right information, which makes them worried.

The old way of doing things does not let the nursery staff watch the kids in time, so they do not know if the kids are playing or sleeping. They also do not have a way to automatically check on the kids.

Because of this some important things may be. The parents and the nursery may not talk to each other as much as they should.

So, we want to make a system that can tell us which kid is which lets us watch them and send messages to the parents and the nursery staff right away. We want to make the nursery a safer place and help the parents and the nursery staff talk to each other.

1.2 Project and Design Objectives

The project aims to design and develop a smart child monitoring system that allows parents to view their children in real time and be assured of their safety throughout the day.

- This will provide a reliable and simple way to identify each child using identification methods such as NFC tags or stickers.
- The system incorporates a child presence sensor on the seat that activates an alarm when the child sits down and requests a scan using NFC or similar identification technology to automatically identify and verify the child's identity.
- Real-time status and activity notification of children will be sent to their parents.
- Developing a mobile application for parents to monitor their children remotely and get real time updates.
- Integration of hardware components, such as sensors and a camera, to enhance the experience.
- The child's inactivity for a set amount of time will be regarded as sleep and will be detected to notify the nursery administration automatically.
- The project will be developed to be reliable, user-friendly, and guarantee continuous communication between the stroller and the parents.
- Activating an automatic wheel locking mechanism when the vehicle approaches potential danger sources to enhance safety and prevent accidents.

1.3 Intended Outcomes and Deliverables

By the conclusion of this project, an intelligent system will be created that allows parents and nursery personnel to observe children with a stroller fitted with a camera and sensors. This system will depend on sophisticated technologies to gather information regarding the child's health and behavior. To improve this system, a mobile app will be created enabling remote monitors to get alerts regarding the child's activities

1.4 Summary of Report Structure

Chapter 2 presents the project's background. Chapter 3 explains the system design in general. Chapter 4 presents and discusses the results. Chapter 5 details the resources used in the project. Chapter 6 discusses the impact of the engineering solution. Chapter 7 presents the conclusions and recommendations.

2 CHAPTER2: BACKGROUND

2.1 Overview

Child monitoring systems are crucial in areas where safety and prompt assistance are essential. In nurseries it's extra important to keep an eye on kids all the time because they might not be able to tell you if they're in trouble. Traditional ways of watching kids rely on people. When staff are looking after lots of children at once, they might not notice something right away.

Our project is trying to fix this problem by suggesting a system that uses the Internet of Things, sensors, cameras, mobile apps, and special tech to identify each child. This system is different because it puts the child at the center of everything rather than just tracking the stroller.

It does lots of things all in one go:

- It identifies the child using tags
- It keeps an eye on the child's state with sensors and a camera
- It sends alerts in time
- It spots if the child is inactive which might mean they're sleeping,
- It locks the wheels automatically if there’s a risk.

This makes the project great for nurseries, where it’s crucial to keep an eye on each child, send alerts, and talk to both parents and nursery staff.

The child monitoring system is really helpful in these situations.

It uses child identification technology to keep track of each child.

The system sends alerts to parents and nursery administration.

The child monitoring system is a tool for keeping kids safe.

2.2 Related Work

Several studies have explored the use of Internet of Things (IoT), sensing technologies, mobile applications, and identification systems to improve child monitoring and safety. These studies provide useful insights for the design of the proposed smart child monitoring system.

2.2.1 USING INTERNET OF THINGS FOR CHILD CARE: A SYSTEMATIC REVIEW

Saeedbakhsh *et al.* reviewed the use of IoT technologies in childcare and highlighted the role of wearable sensors, tracking systems, RFID/NFC technologies, and cloud-based platforms in monitoring children and improving safety [2]. The review showed that IoT-based childcare systems are mainly designed to collect real-time information about the child's movement, activity, and surrounding conditions, while allowing parents and caregivers to access this information remotely. As Figure 2.1 presents the distribution of study environments (indoor, outdoor, both, and unKnown), providing an overview of the typical deployment settings of IoT-based childcare systems.

This study is relevant to the proposed project because both systems rely on IoT technologies for real-time child monitoring and aim to improve safety through sensor-based observation and communication. However, the reviewed systems mainly focus on monitoring and sending alerts, while the proposed project extends this concept by integrating a camera, NFC-based identification, sleep detection through inactivity, and an automatic wheel-locking mechanism to provide a more active safety response.

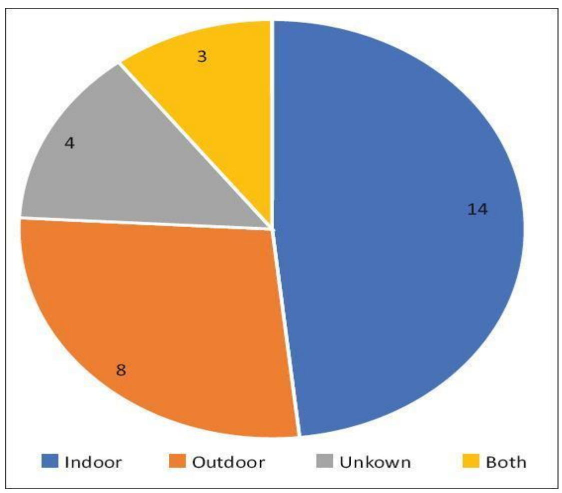


Figure 2-1 Study environments (indoor, outdoor, Both, UnKown)

2.2.2 DESIGN OF AN INTELLIGENT STROLLER CONTROL SYSTEM BASED ON THE INTERNET OF THINGS

Yang and Li proposed an intelligent stroller control system based on IoT technology, wireless communication, and sensor integration [3]. As shown in Figure 2.2, Their system uses a microcontroller together with sensors such as obstacle-detection sensors, speed sensors, and vibration sensors, and it connects to a mobile application through wireless networking. The system is designed to monitor stroller movement, detect abnormal situations, and send notifications to the user.

This work is similar to the proposed project because both systems use IoT architecture, a microcontroller, multiple sensors, and mobile-application connectivity to improve safety. However, the main focus of the reference system is monitoring and controlling the stroller itself, while the proposed project focuses on monitoring the infant. In addition, the proposed project includes NFC-based child identification, live camera support, inactivity-based sleep detection, and direct communication with both parents and nursery administration, which makes it more suitable for nursery use.

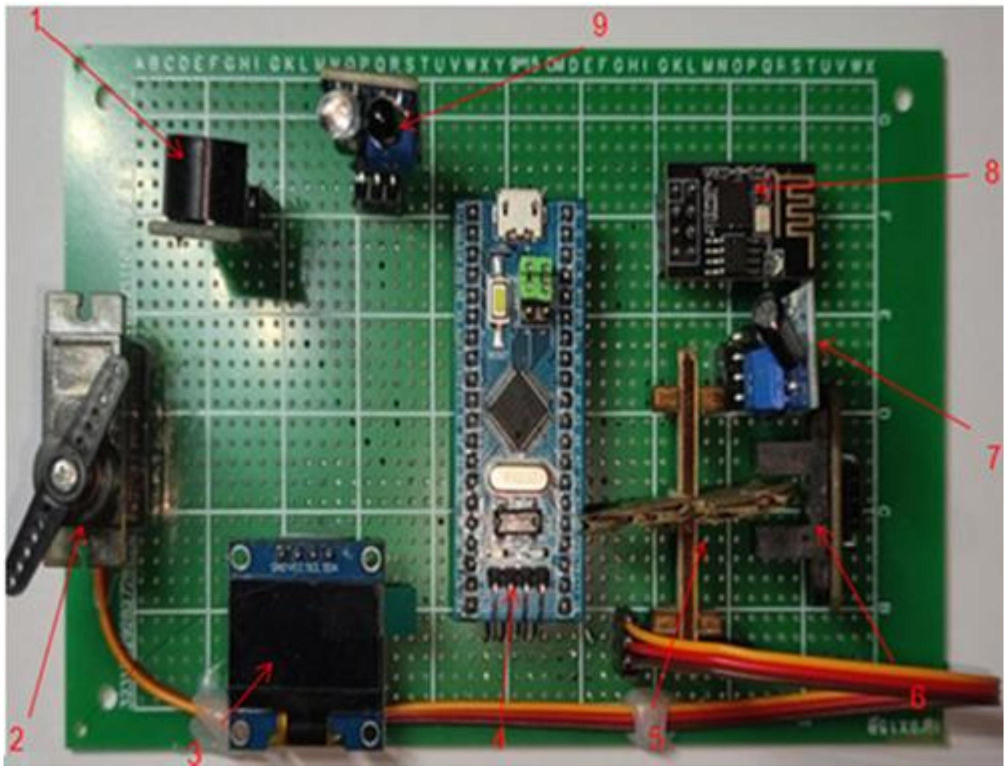


Figure 2-2 shows the system components, including the STM32 microcontroller, Wi-Fi module, display, motor, and various sensors.

2.2.3 IOT BASED SMART BABY MONITORING SYSTEM WITH EMOTION RECOGNITION USING MACHINE LEARNING

Alam *et al.* developed an IoT-based smart baby monitoring system that combines sensors, a camera, and machine-learning techniques for real-time monitoring of the child's condition [4]. As shown in Figure 2.3. Their system supports camera-based observation, live streaming, and mobile-application alerts, and it also allows control of certain system functions remotely.

This research is closely related to the proposed project because both systems use IoT technologies, multiple sensors, camera-based monitoring, and real-time alerts through a mobile application. However, the reference system does not support NFC-based child identification and is mainly limited to communication with parents only. It also does not involve nursery administration or provide a multi-user nursery-oriented design. In addition, it lacks a dedicated feature for detecting sleep through inactivity analysis, whereas the proposed project includes these capabilities together with an automatic wheel-locking mechanism for additional safety.

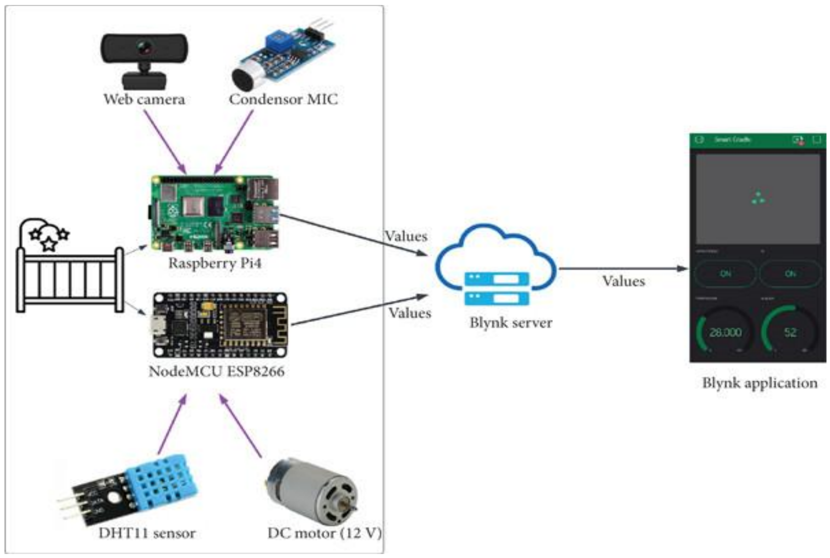


Figure 2-3 System architecture.

2.2.4 DEVELOPING A SMART NURSERY APPLICATION FOR MONITORING AND BABIES CARE

Alqahtani *et al.* presented a smart nursery application that allows parents or caregivers to monitor children remotely through a connected application interface [5]. The system provides monitoring functions such as child information display, application-based access, and camera-based observation of activity. The use case diagram in Figure 2.4 shows how users (parents, supervisors, and technical support) interact with the smart nursery system and its main functions such as login and child monitoring.

This study is relevant because it addresses nursery monitoring and application design, both of which are important components of the proposed project. It also provides a useful model for organizing child information and displaying monitoring data to users. However, the reference system mainly depends on a general surveillance camera for monitoring multiple children, while the proposed project depends on individual monitoring for each child using dedicated sensors and identification technology. This gives the proposed system higher privacy, better precision, and more direct automatic alerting without requiring continuous video observation.

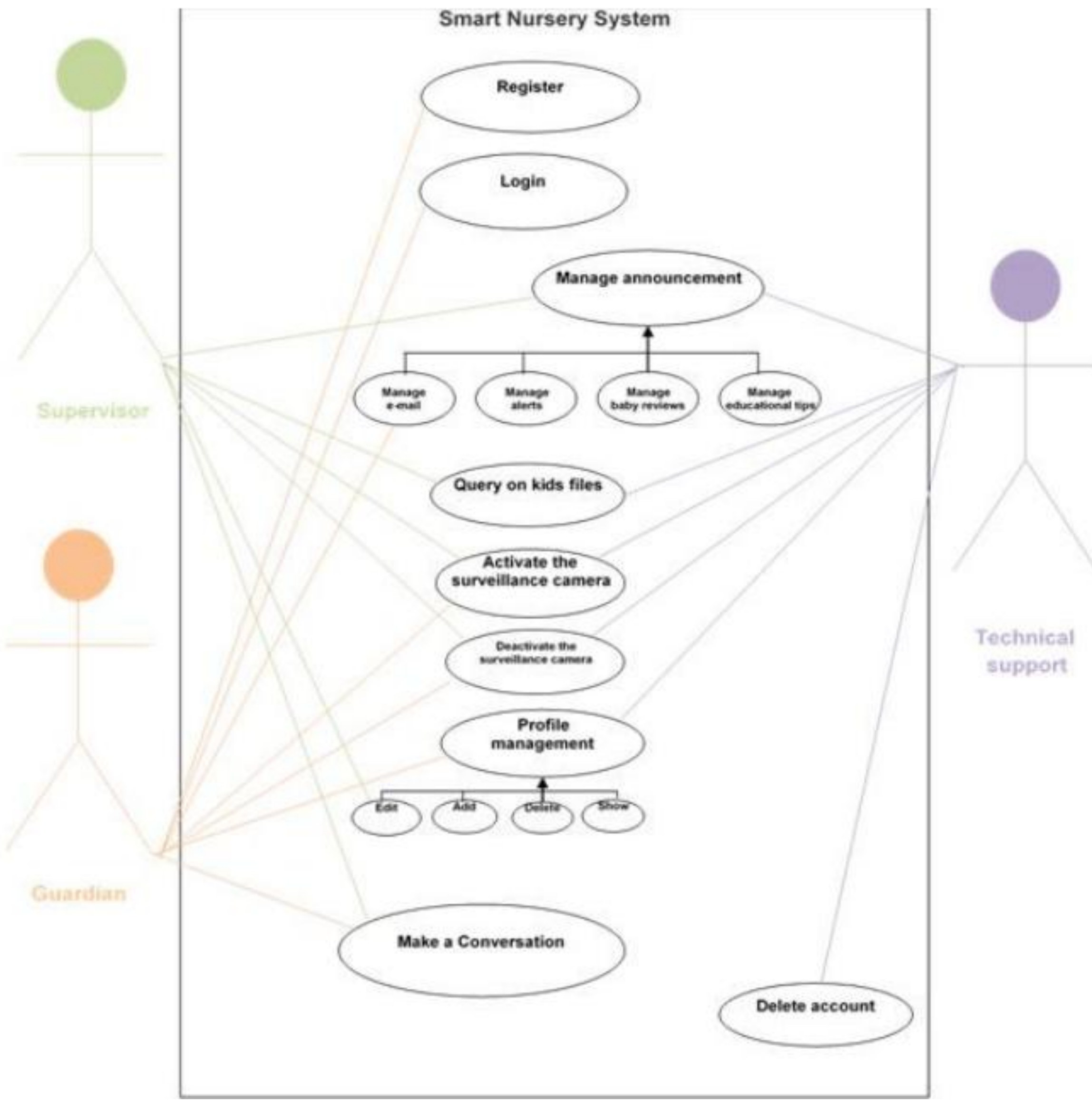


Figure 2-4 The use case diagram.

2.2.5 CUSTOM MADE NFC TAGS FOR NEWLY BORN BABIES AT HOSPITAL

Iqbal *et al.* proposed the use of custom NFC tags for newborn babies in hospitals to simplify identification and reduce errors associated with manual registration [6]. AS shown in Figure 2.5, the baby's information is stored in an NFC tag and can be retrieved using an NFC-enabled device.

This work is highly relevant to the proposed project because it demonstrates the usefulness of NFC technology in securely linking child identity data with a digital system. The proposed project can benefit from this concept by using NFC to identify the child and connect the child to the monitoring platform. However, the hospital-based NFC system mainly focuses on identification and information retrieval through repeated scanning, while the proposed project uses NFC only as the initial linking step. After that, continuous monitoring is performed automatically using sensors and a camera without requiring repeated scans.

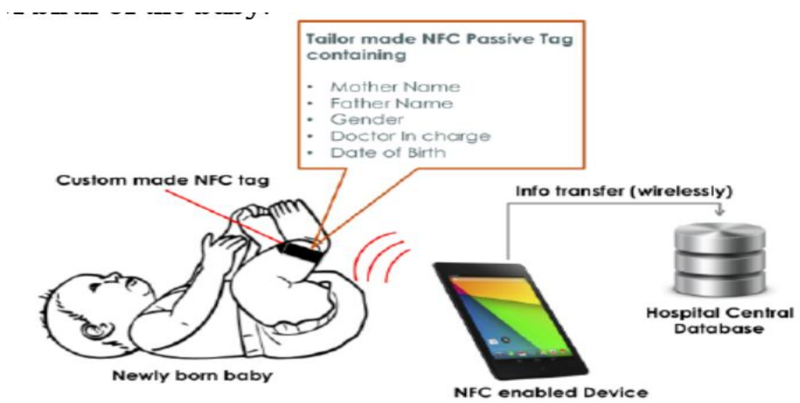


Figure 2-5 NFC tags based system for newly born babies at hospital

2.2.6 DEVELOPMENT OF A BABY WALKER WITH AN INTEGRATED SPEED-LIMITING SYSTEM TO ENHANCE SAFETY AND SUPPORT INFANT MOTOR SKILL DEVELOPMENT

Islamiyati *et al.* presented a baby walker system designed to improve infant safety by integrating a speed-limiting mechanism into the walker [7]. The system uses sensing and control concepts to monitor movement and reduce accident risk during walker use. It is intended to create a safer walking environment for infants while also supporting motor-skill development. Figures 2.6 and 2.7 show baby walker designs and types used for safety purposes.

This study is related to the proposed project because both systems aim to improve child safety during movement and both rely on IoT-based sensing and control concepts. In both cases, the system uses a technical safety mechanism that can intervene when unsafe movement or danger is detected. However, the reference system mainly focuses on speed control and safe walker movement, while the proposed project provides a complete child monitoring system. The proposed system includes live camera monitoring, NFC-based identification, a mobile application for real-time updates, inactivity-based sleep detection, and an automatic wheel-locking mechanism for use in nursery environments.

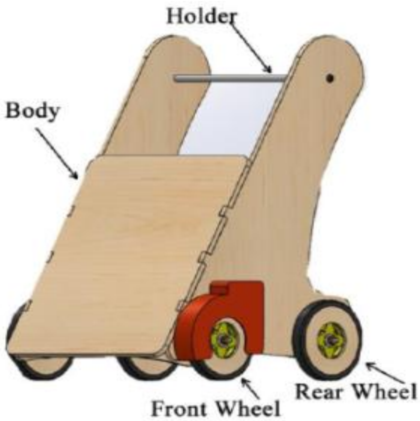


Figure 2-6 A prototype of the baby walker

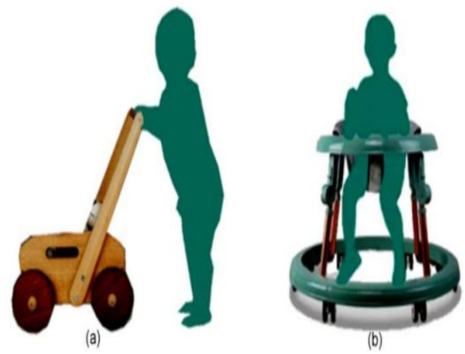


Figure 2-7 Types of baby walkers available

2.2.7 A SMART WALKER FOR PEOPLE WITH BOTH VISUAL AND MOBILITY IMPAIRMENT

Mostofa *et al.* proposed a smart walker designed for users with both visual and mobility impairments [8]. As shown in Figure 2.8, the system investigates sensing technologies, real-time data processing, and intelligent safety mechanisms to improve obstacle awareness and reduce movement-related risks.

Although this work is not specifically about child monitoring, it is useful because it demonstrates how sensors, real-time processing, and safety-oriented control mechanisms can be integrated into a walker-based platform. This is related to the proposed project in terms of obstacle awareness, automated response, and movement safety. However, the reference system is designed for assistive walking for visually and mobility-impaired users, whereas the proposed project is designed for child monitoring and protection in nurseries.

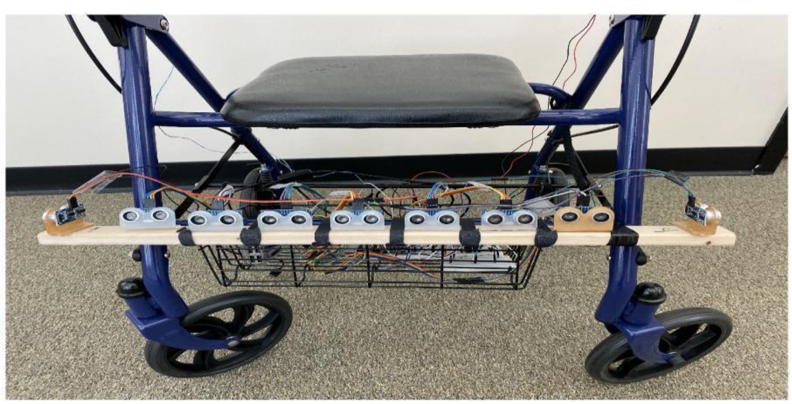


Figure 2-8 The rollator uses ultrasonic sensors for obstacle detection and navigation

2.3 Comparison Between Previous Studies and the Proposed Project

To better position the proposed project within the existing literature, this section compares the most relevant previous studies with the proposed smart child monitoring system. The comparison focuses on the main similarities and differences between the reviewed works and the proposed project in terms of monitoring approach, sensing technologies, identification methods, safety mechanisms, and application support.

	NFC recognition	Sensing the presence of the child	Alerts	mobile app	Camera and Sensors	Sleep detection	communication	Wheel lock
Using Internet of Things for Child Care: A Systematic Review	RFID/NFC is used for general tracking, not for precise individual identification.	Not specified	Alerts are based only on sensor readings.	Remote parental data viewing is possible.	Relies on multiple sensors without camera focus.	Does not support sleep detection or motion detection.	Good communication with parents and supervisors.	Provides alert only, no direct action
Design of an Intelligent Stroller Control System Based on the Internet of Things	There is no definition of a child, the focus is only on the stroller.	No child detection	Motion and obstacle alerts	App for stroller control and tracking	Powerful motion and obstacle sensors without a clear camera	No child status analysis	Limited user interaction	Motion control only, no full lock
IoT Based Smart Baby Monitoring System with Emotion Recognition Using Machine Learning	It does not have NFC or child identification	No seat sensor	Smart alerts via camera and analysis	App displays live video and alerts	Combines camera and sensors effectively	Does not rely on motion to detect sleep	Communicates only with parents	No mechanical controls
Developing a Smart Nursery Application for Monitoring	None	None	App-only alerts	Entirely app-based	Relies only on the camera with no additional sensors	None	Good communication between parents and the system	None

	NFC recognition	Sensing the presence of the child	Alerts	mobile app	Camera and Sensors	Sleep detection	communication	Wheel lock
and Babies Care								
Custom Made NFC Tags for Newly Born Babies at Hospital	Very strong NFC recognition, but only when scanning	None	Does not provide instant alerts	Very limited interface	No sensors or camera	None	Simple communication for data displays only.	None
Development of a Baby Walker with an Integrated Speed-Limiting System to Enhance Safety and Support Infant Motor Skill Development	None	None	No live alerts, only speed reduction	None	Motion and speed sensors only	None	None	Provides partial safety by reducing speed only
A Smart Walker for People with Both Visual and Mobility Impairment	None	None	Obstacle alert only	None	Powerful obstacle detection sensors	None	None	No lock or full control
KidSentinel	It is used to identify the child	It is designed to detect the presence of the child in the seat.	Real-time alerts for sleep	provides real-time monitoring and notifications	Provides camera-based monitoring for each stroller.	Detects sleep based on inactivity over a period of time.	Sends notifications to the nursery and allows parents to monitor the child remotely.	It locks the wheels when approaching danger

Table 2-1 Comparison Table.

It has already been proved in previous research that the use of Internet of Things, sensors, and smart applications is very important for child monitoring. But most of the existing applications concentrate their attention on individual operations like child monitoring, recognition, or movement control. Our project combines all these functions in a single system that gives the opportunity to monitor and stream videos from the stroller in real time.

3 CHAPTER 3: METHODS AND MATERIALS

3.1 System Design and Components

3.1.1 SYSTEM DESIGN

The block diagram of the KidSentinel System is presented in Figure 3.1 below. It shows the composition of the intelligent child monitoring system with hardware devices that interact with the Firebase. The ESP32 microcontroller is the central part of the system that unites all other sensors and works as their connector and data analyzer. The FSR sensor detects whether the child is present in the seat. Besides, the MPU6050 sensor monitors the baby's movements, and the ultrasonic sensor monitors obstacles.

NFC Reader identifies the child with the help of his card data, and the information is sent to the microcontroller for further processing. The system makes numerous decisions according to the situation. Thus, when a threat occurs, it will sound the alarm or notice the baby sleeping by the absence of his movement during a particular time. Moreover, a servo motor will be engaged to stop the stroller in case of a detected threat.

ESP32-CAM will make it possible to monitor the child with its camera function. All the obtained data, for example, that the child was identified, he sleeps, or there is a threat will be sent to the Firebase. It will inform the parents or daycare representatives about these facts via the corresponding mobile application.

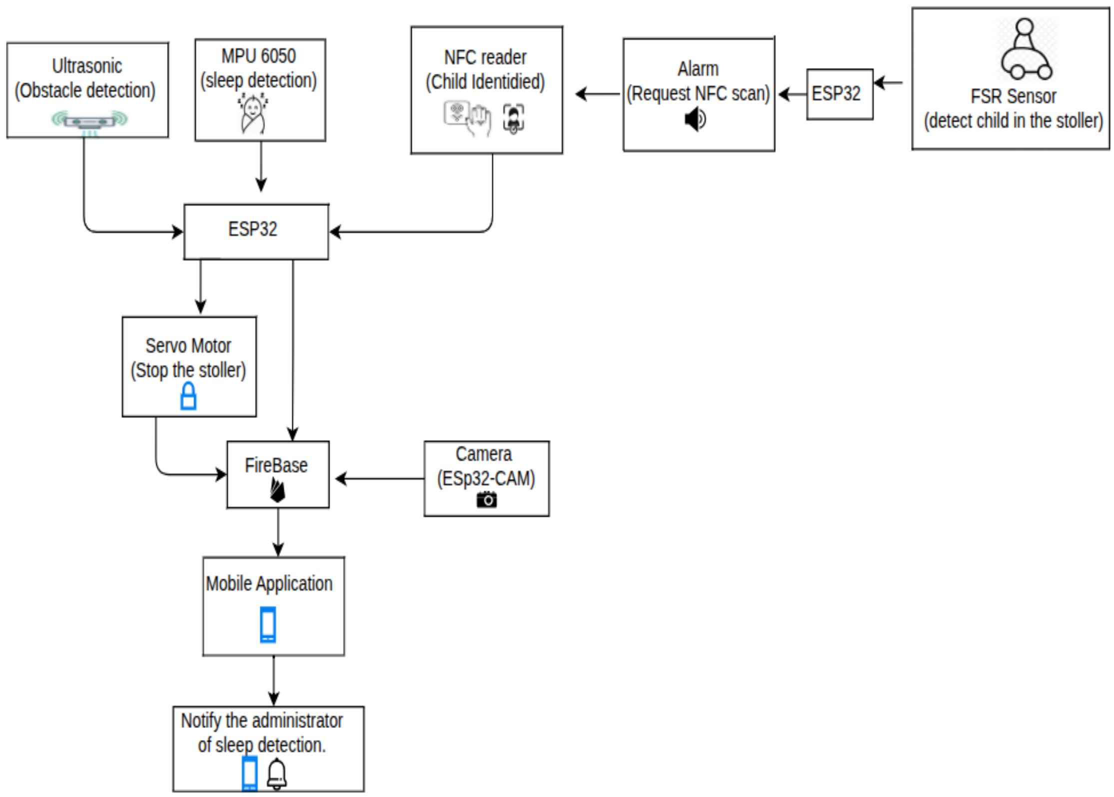


Figure 3-1 Block Diagram

3.1.2 HARDWARE COMPONENTS

In this section, we will explain the hardware components that we will use in our Project.

HW component:

- MPU-6050
- FSR sensor
- Servo Motor
- NFC
- Buzzer
- Ultrasonic sensor
- ESP32
- ESP32-CAM Camera Module

- Battery
- Wires

1) MPU-6050:

The miniature device comprises a 3D gyroscope and a 3D accelerometer that measure different parameters such as acceleration, rotation, and orientation, as shown in Figure 3.2 and Figure 3.3.

Sensor information is measured using the I2C data interface and relayed to the microcontroller [9]. The input voltage required for operation by the device is from 2.375 to 3.46V while the module is capable of working between 3 to 5V [10]. The device further uses a 16 bit ADC converter to accurately measure movement.

This project, the MPU6050 sensor is used to detect whether the baby is sleeping or awake. The sensor measures any form of acceleration or movement. Movement is associated with being awake, while the lack of movement means that the baby is asleep.

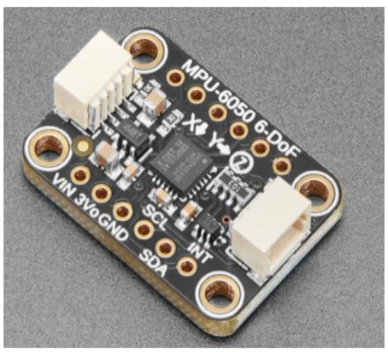


Figure 3-2 MPU-6050

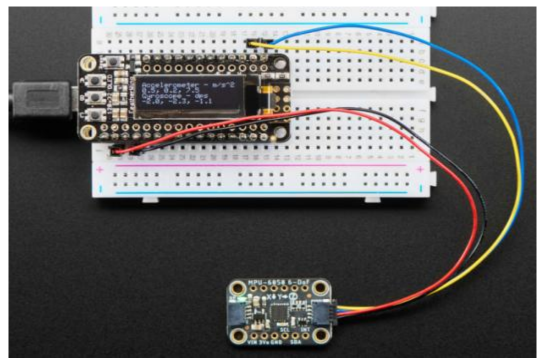


Figure 3-3 MPU-6050

2) FSR sensor

FSR serves as a pressure-dependent resistor where the sensor features high resistance when there is no pressure applied and low resistance when the pressure rises. As the sensor itself does not produce a voltage output, the voltage divider circuit with a constant resistor can be used to measure the resistance level which will provide the microcontrollers (e.g.,ESP32 or Arduino) with a voltage value (As shown in Figures 3.4-3.6). Then the voltage levels will be digitized to define whether the pressure is applied or not [11].

This project, the FSR sensor detects whether the kid is on a chair or not. In other words, when the kid uses a chair, the pressure levels change, thus producing new values. Based on the new values obtained from the sensor, ESP32 detects the presence of the kid and generates an audible signal.

3) Servo Motor

Figure 3-4 FSR sensor

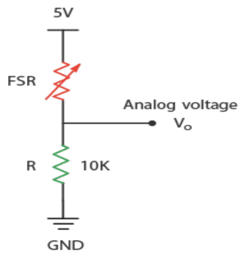


Figure 3-5 Reading an FSR

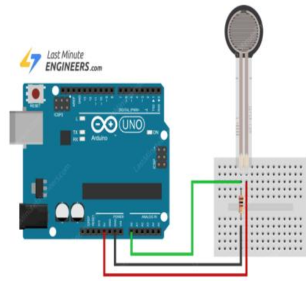


Figure 3-6 FSR connection

Servo motor acts within the closed loop system of control where feedback system ensures the exact measurement of angle of the motor. A pulse-width modulation signal representing the desired angle is provided by the microcontroller. The difference between actual and desired angles is calculated by comparing the current signal with the signal from the internal sensor like potentiometer/encoder. In case there is a difference between two signals, the motor starts moving in order to reduce the difference between two angles until it reaches the desired angle. There are kinds of servo motors. We have AC servo motors and DC servo motors as shown in figure 3.7. Each servo motor is used for things depending on what the application needs [12].

This project, DC servo motor will use for creating an obstacle detecting wheel-locking mechanism. Servo motor uses a PWM signal received from ESP32 in order to calculate the angle of movement. In the presence of obstacles, servo motor assumes the locking position for starting the braking mechanism while in normal condition it resumes its initial position.

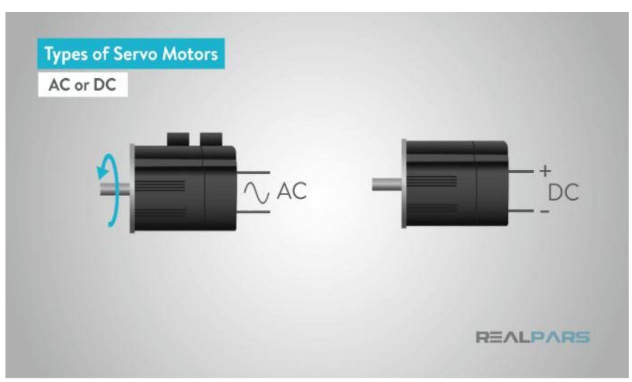


Figure 3-7 Types of Servo Motor

4) NFC Tag and NFC Reader

NFC used to enable the interconnection of devices placed closely together [13]. In our case, we use the NFC technology in conjunction with NFC RC522 reader and ESP32 microcontroller. As shown in Figure 3.9, NFC Reader is used in the system. When the baby is located in the stroller, the NFC system detects the code on the tag used by the child. It becomes a way to relate each child to their profile in the system. This is how the system ensures safety by preventing unauthorized access.

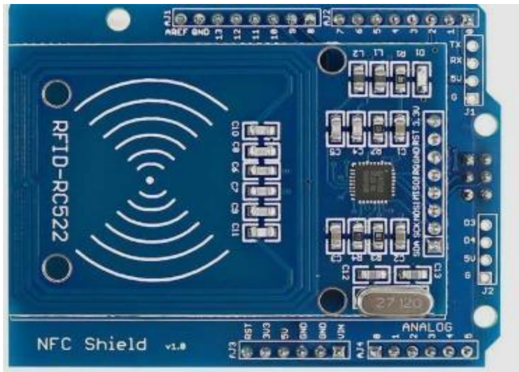


Figure 3-8 NFC Reader

Figure 3-9 NFC Tags

5) **Buzzer**

The buzzer makes a sound when it senses a child in the stroller to alert the user to perform an NFC scan.

Figure 3-10 Buzzer

6) **Ultrasonic Sensor**

An ultrasonic sensor is used to detect obstacles. This sensor measures distance by sending an ultrasonic pulse and measuring the time it takes for the echo to return. This sensor supports non-contact distance measurement within a common range of approximately 2 cm to 400 cm [14].

In the KidSentinel system, the ultrasonic sensor is placed at the front of the walker or stroller to detect nearby objects. The ESP32 processor reads the measured distance and compares it to a predefined safe distance threshold. If the detected distance is less than the safe threshold, the system considers it a potential hazard. In this case, the ESP32 processor updates the Firebase database with the warning status and sends a control signal to the servo motor to activate the stop mechanism.

Figure 3-11 Ultrasonic Sensor

7) ESP32 Microcontroller

The ESP32 is selected as the main microcontroller because it provides built-in Wi-Fi and Bluetooth capabilities, multiple GPIO pins, analog inputs, PWM outputs, and support for communication protocols such as I2C, SPI, and UART. The ESP32-WROOM-32 module is designed for IoT applications, including sensor hubs, data loggers, and camera/video-related systems [15]. These features make it suitable for connecting the different sensors and output components used in the KidSentinel system.

In this project, the ESP32 receives sensor readings from the FSR sensor, MPU6050 motion sensor, ultrasonic sensor, and NFC reader. It processes these readings and controls the output devices, such as the buzzer and servo motor. The ESP32 also communicates with Firebase through Wi-Fi, allowing real-time system data to be viewed through the mobile application.

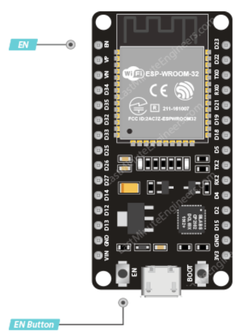


Figure 3-12 ESP32 Microcontroller

8) ESP32-CAM Camera Module

The ESP32-CAM is a compact camera module based on the ESP32 platform. It supports Wi-Fi communication and usually includes an OV2640 camera module, which makes it suitable for IoT monitoring and visual surveillance applications [16].

In the KidSentinel system, the ESP32-CAM is used to provide visual monitoring of the child. This supports parents or nursery administrators by allowing them to check the child’s condition when needed. The camera is especially useful when sensor readings alone are not enough to understand the situation, such as when inactivity is detected and the system classifies the child as sleeping.

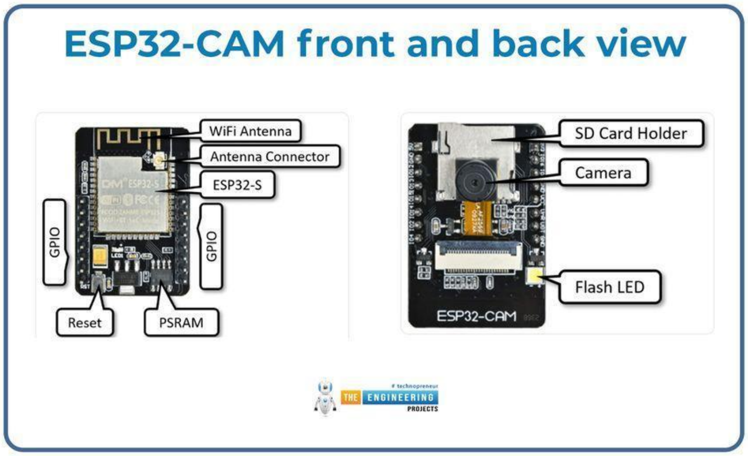


Figure 3-13 ESP32-CAM Camera Module

9) Power Bank

A rechargeable USB power bank was selected as the main power source for the KidSentinel prototype. Because the system is installed on a stroller and must operate as a portable unit without depending on a fixed wall outlet. A power bank is rechargeable, replaceable, easy to connect, and safer for prototype implementation compared with exposed battery cells.

The power bank should provide a stable 5V output with enough current to operate the ESP32, ESP32-CAM, sensors, buzzer, and servo motor. A current rating of at least 2A is recommended, especially when the ESP32-CAM and servo motor operate at the same time. The ESP32 can be powered through the USB port or 5V/VIN pin, while the servo motor should be powered from a suitable 5V supply because it may require higher current during movement. A common ground must be connected between all components to ensure correct operation.

Using a commercial power bank also improves safety because it usually includes protection against overcharging, short circuits, overload, and overheating. For this project, the power bank should be fixed securely inside the enclosure to avoid disconnection or damage during nursery use [17].

Figure 3-14 Power Bank Unit

3.1.3 SOFTWARE COMPONENTS

Our project includes a set of software components that work together to build a mobile application in order to operate the system in an integrated manner, linking the sensors and control unit with the mobile application, so that data is exchanged in real-time.

Arduino IDE

The Arduino IDE was used as the main development tool for coding and uploading code onto the ESP32. This platform was used to retrieve data from the sensors, as well as control the servo motor using PWM, which helps in determining the angle of motion accurately. Besides, the communication channel between ESP32 and the cloud was set up for data transmission.

Firebase

The system uses Firebase as a cloud database for facilitating real-time communication between the project modules and the mobile app. Data from the sensors is obtained by the ESP32 and is sent to the Firebase Realtime Database in the form of JSON. The mobile app will have access to this data and use it to constantly monitor the status of the system while managing the child and the stroller at the same time.

Furthermore, Firestore serves as another cloud database that stores crucial data related to the system in an organized fashion. Such data includes information about the parents, children, administrative accounts, as well as information related to strollers or units in the project.

Mobile Application

The program will be built using the programming language called Dart together with the Flutter framework and will then be connected to a Firebase database for efficient data transfer.

3.1.4 SOFTWARE DESIGN

This section showcases the design of the application developed for the project, which aims to monitor the child's condition and ensure their safety inside the smart cart, as well as to help parents and officials receive instant notifications and updates through the mobile application.

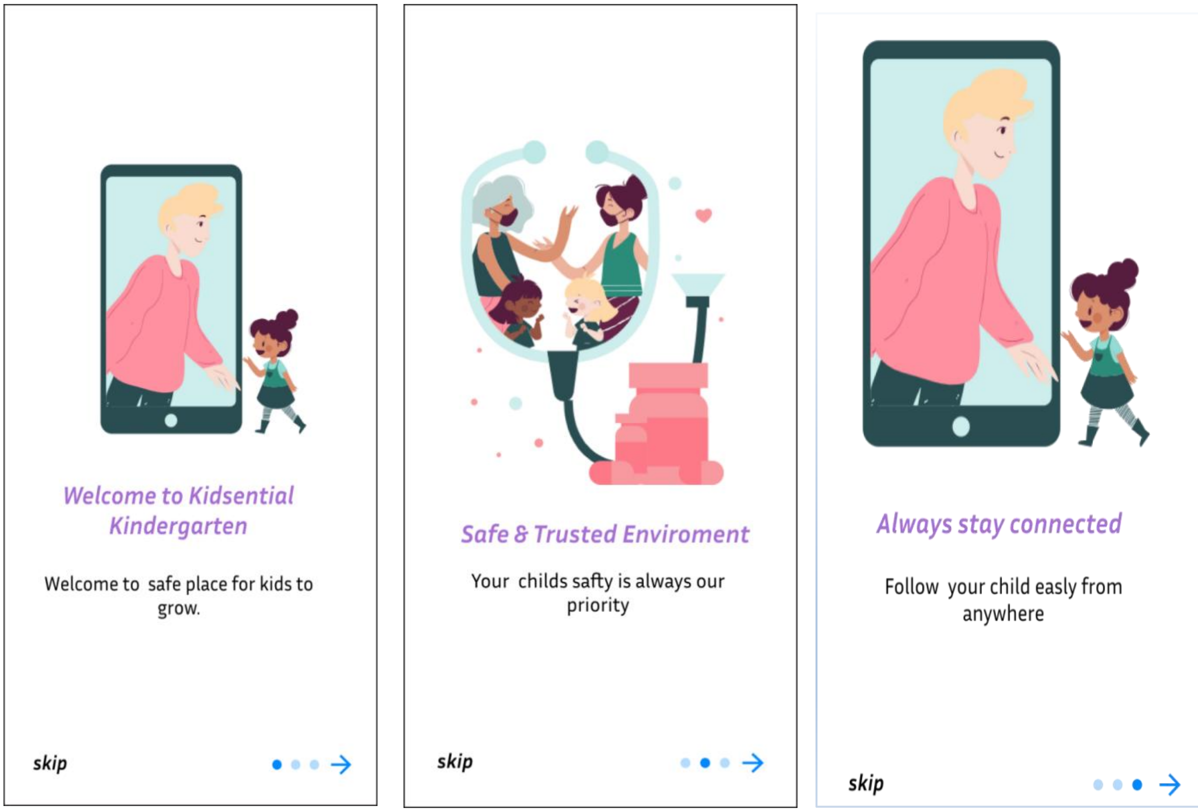


Figure 3-15 Introduction of the app.

After the introductory pages, the user can log in to the application and begin using it. Before logging in, the user must select their account type: Administrator or Parent as shown in figure 3.16. If the user does not already have an account, they can create a new one by entering their personal information, as shown in Figure 3.17.

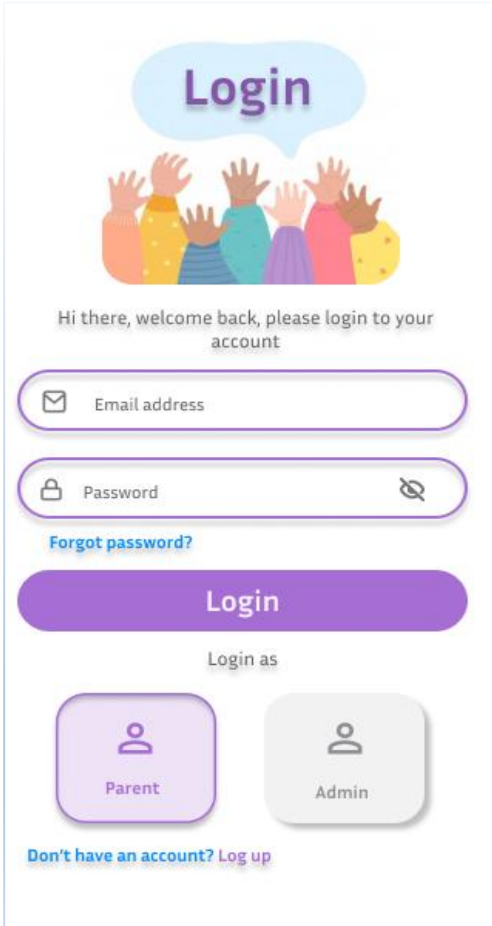


Figure 3-16 Login Page

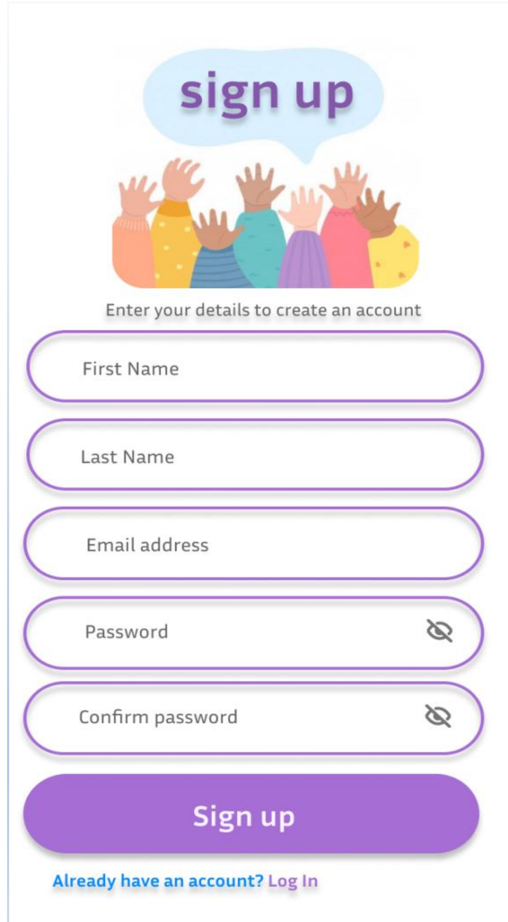


Figure 3-17 Sign up page

Parent Interface consists of three main pages including Home, Camera, and Profile. The Home page (as shown in figure 3.18) provides personal status of the child showing whether he/she is sleeping or awake, and at the same time shows the status of the connection to the NFC chip. In case there is no connection to the NFC chip, the personal data of the child is not shown in order to make sure that the accuracy of the information is maintained. Camera page (as shown in figure 3.19) allows the parent to watch the real-time video of the child using zoom features for better visibility. The Profile page contains account information and data for both the user and the child, as illustrated in Figures 3.20.

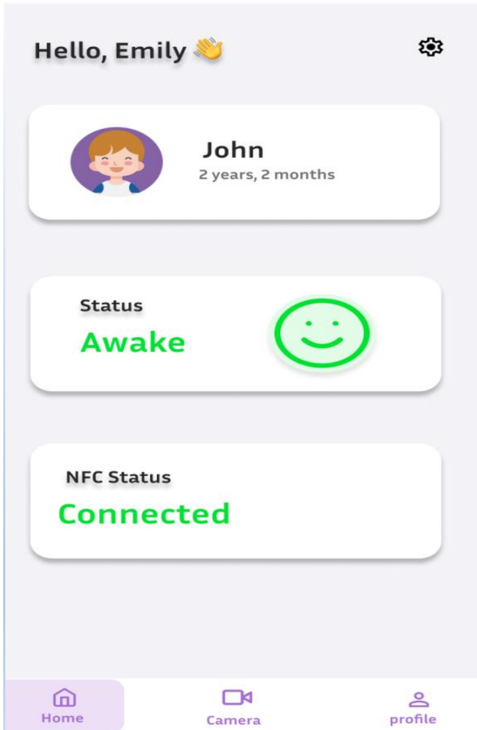


Figure 3-18 Home Page of parents

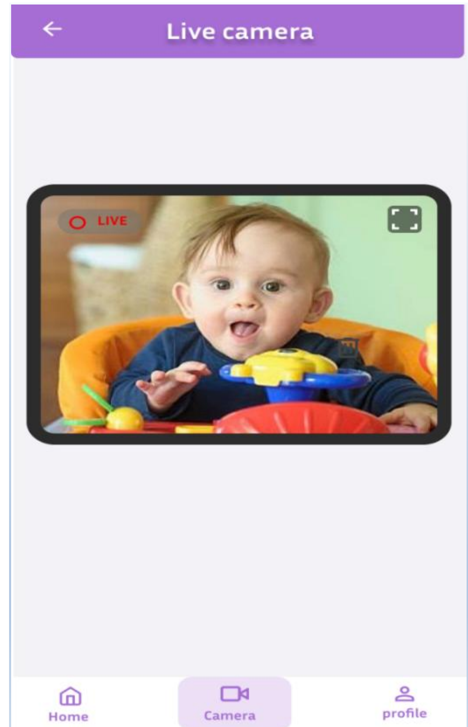
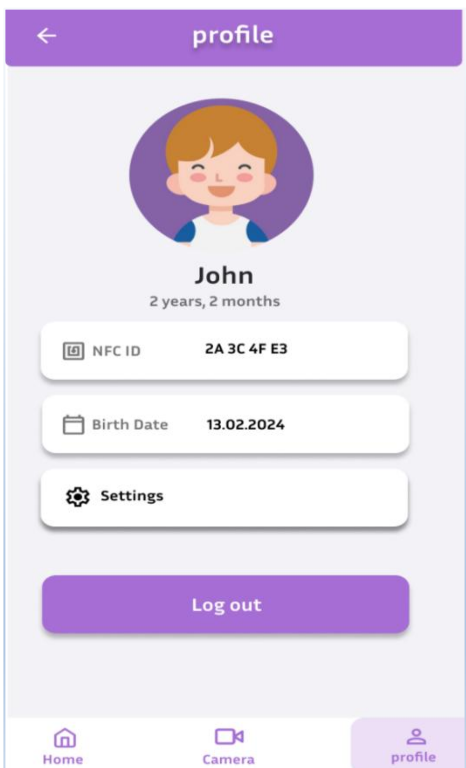


Figure 3-19 Camera page



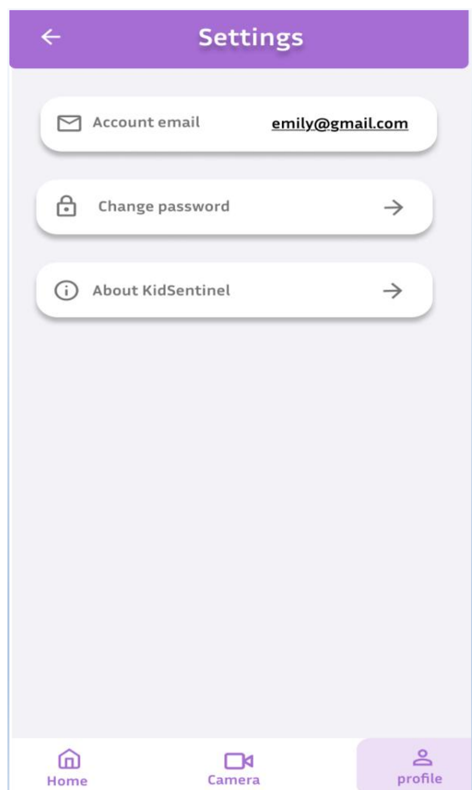


Figure 3-20 Profile and Settings page

There are four major pages in the administrator interface: the Home page, the Children page, the Notifications page, and the Settings page, as depicted in figures 3.21, 3.22, 3.23, and 3.24. The Home page is used for basic navigation in the application. The Children page shows a list of all registered children; clicking on any of them opens the detail page, which shows all information about them. The Notifications page has all notifications raised by children, such as security alerts and connection status. The Settings page lets users add a new child or delete any existing child from the application.

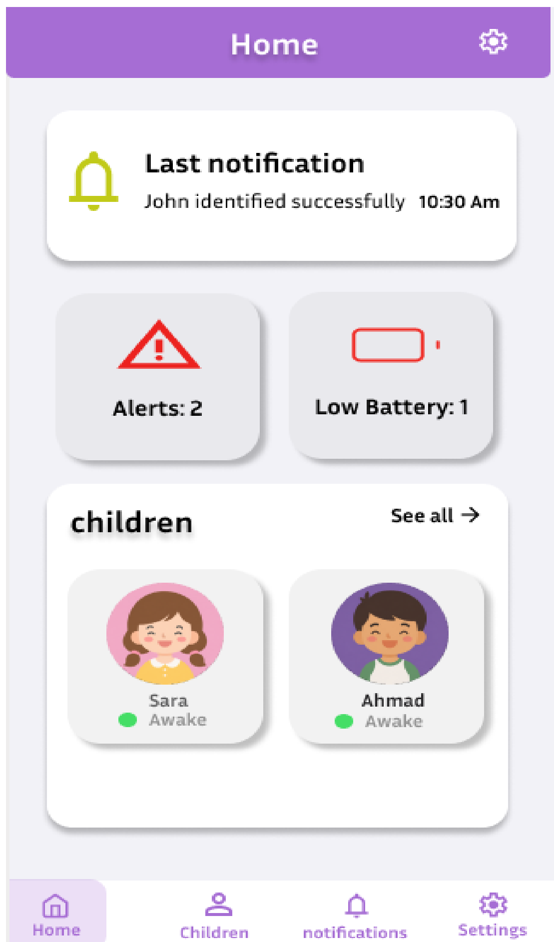


Figure 3-21 Home page

Figure 3-22 Children and Child page

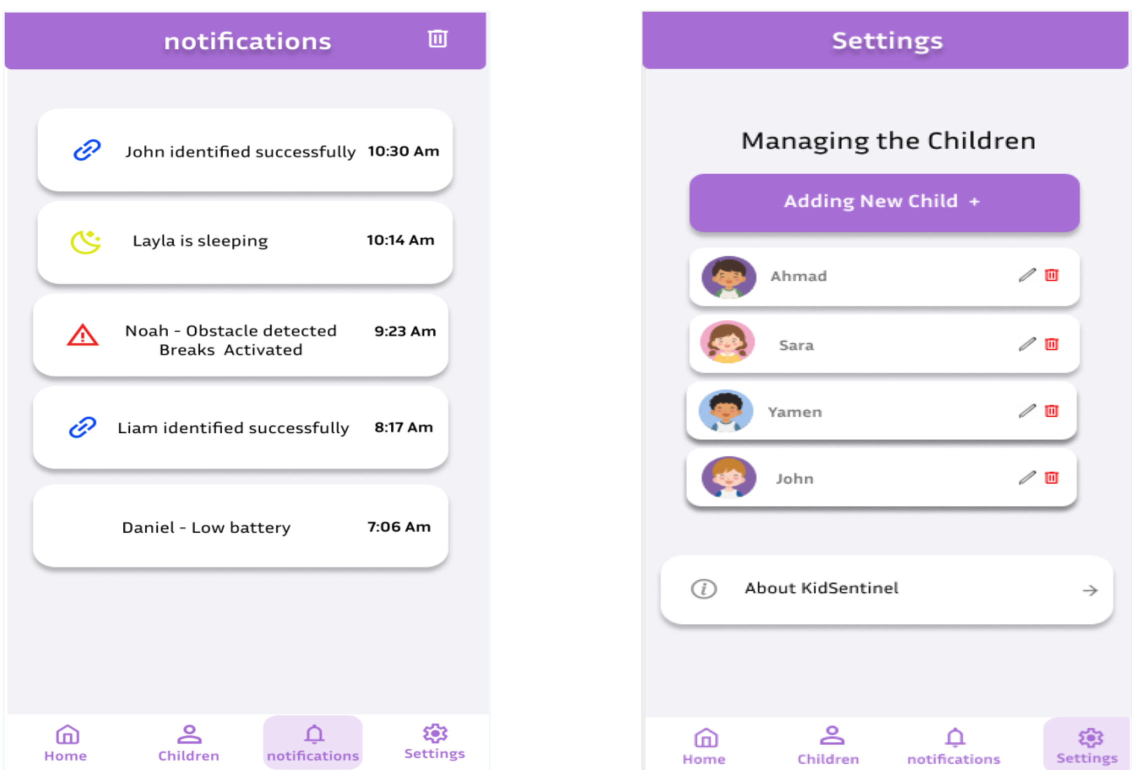


Figure 3-23 Notifications page

Figure 3-24 Settings page

3.2 Design Specifications, Standards and Constraints

This section presents the main specifications and constraints of the KidSentinel system. The specifications describe the required functions of the system, while the constraints describe the limitations that may affect the implementation and performance of the prototype.

3.2.1 SPECIFICATIONS

The following are the specifications that will be utilized for the creation of the KidSentinel system.

- FSR sensor will detect if the child is seated.
- NFC scanning will be requested after detecting the child in the stroller/walker.
- NFC tags will identify the child.
- MPU6050 sensor will monitor the child's movements.
- If the child is not moving for a defined period of time, the system will detect when the child is sleeping.
- Ultrasonic sensor will detect if there are obstacles near the child.
- If any danger is detected, the servo motor will activate to stop the child's walker.
- ESP32-CAM will provide live, visual monitoring of the child.
- Real time data will be sent to Firebase.
- Information will be displayed on the mobile application for the parents and administrators
- Alerts will be sent to the administrator when the child is detected sleeping.
- Access to the information of the child and the monitoring through the camera will be limited to the authorized users only.
- The system checks if the NFC tag has been scanned before allowing access to the monitoring system.

3.2.2 CONSTRAINTS

The KidSentinel system also has some constraints.

- The system requires a Wi-Fi connection to the Firebase database and to send notifications to the mobile devices.
- The ESP32 and ESP32-CAM modules require a stable power source. The power bank has to supply enough current for the system components.

- The power bank will need to be continuously charged or regularly recharged to operate the system.
- The sensor readings could be affected by the placement of the sensors on the child's body, the child's weight, external noise, and other surrounding conditions.
- The FSR sensor, MPU6050 sensor, and the ultrasonic sensor need to be calibrated to reduce false sensor readings.
- The servo motor needs to be powered correctly as it may require more current than the ESP32 can provide.
- As the system will include a camera to monitor the child, privacy concerns will be created. Only the authorized parents and administrators of the nursery will have access to this information.
- All of the system's wires, sensors, and electronic components will have to be secured within a safe enclosure as the system will be in close proximity to the children.

3.3 Design Alternatives

3.4 System Analysis and Optimization

3.4.1 FLOW CHART:

Figure 3.25 illustrates the flow chart of the system. The system begins by turning on all devices while ensuring that there is connectivity. If there is no connectivity, the system continues trying to connect until it finally succeeds. After connecting to the internet, the system remains idle waiting for any child to come. In case there is a child, the system produces a beeping noise, which is necessary for the child to be identified via NFC technology. If the identification is successful, the system starts collecting sensory data in order to monitor the child's condition. If no activities are detected, the system sends a message through the phone app of the administrator. If there is danger in the environment of the stroller, then the system activates the wheel lock.

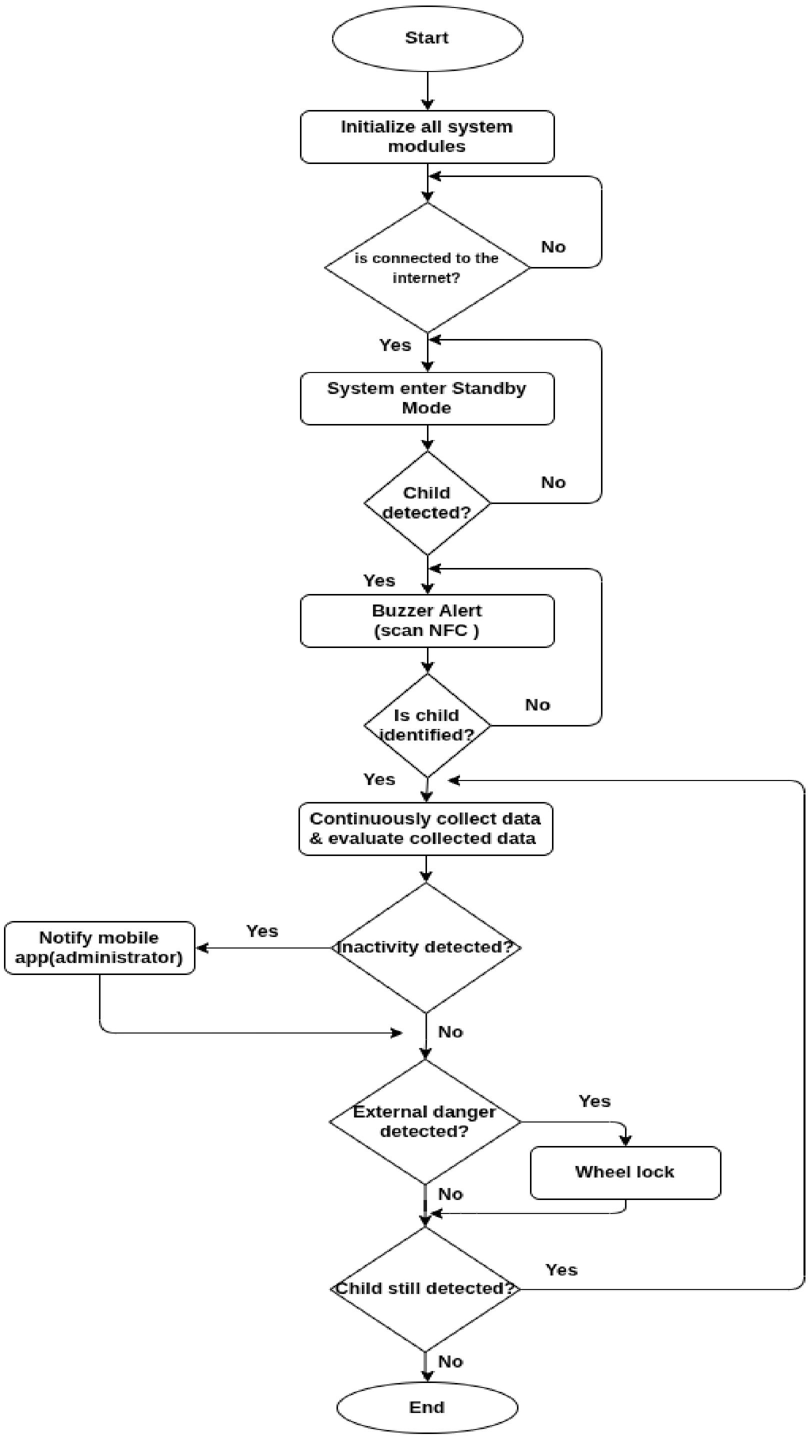


Figure 3-25 Flow Chart of the system

3.4.2 USE CASE:

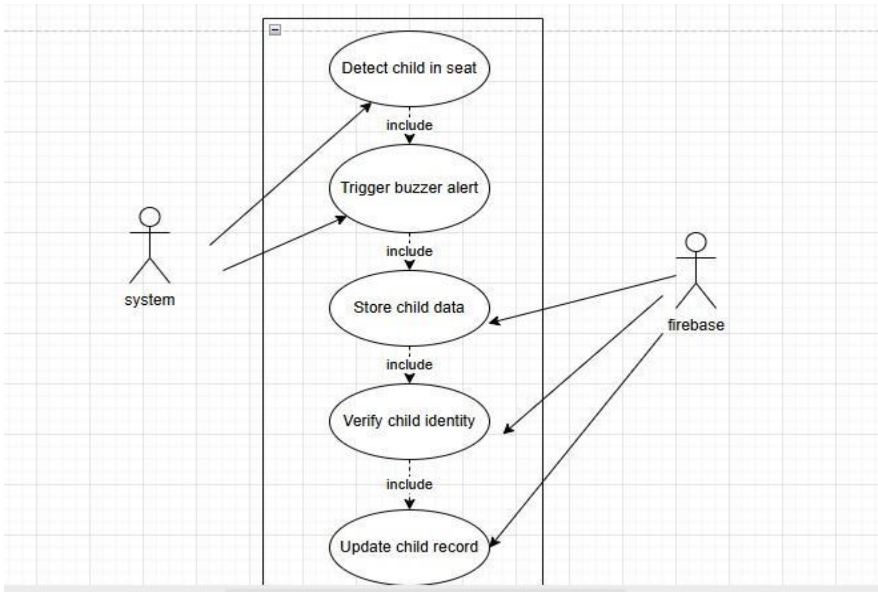


Figure 3-26 Child identification use case

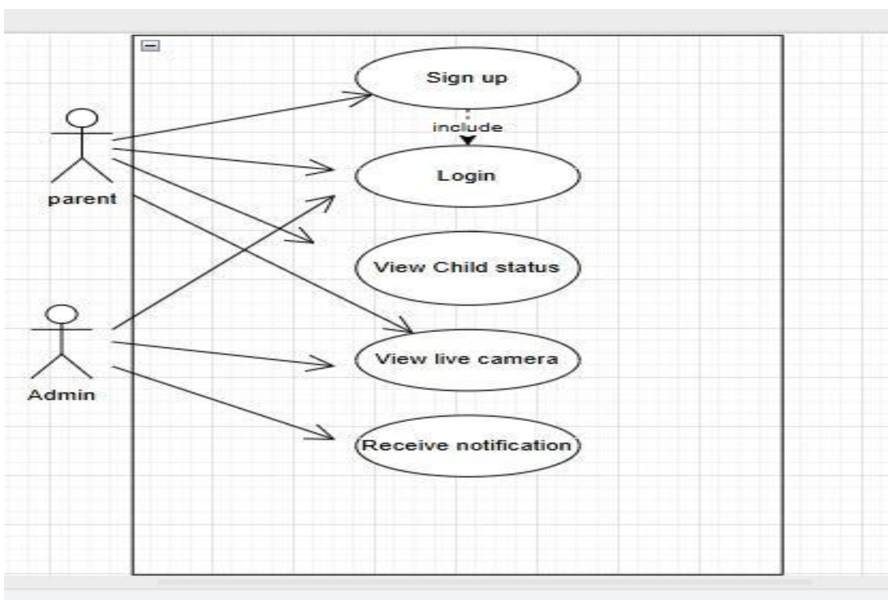


Figure 3-27 Mobile application use case

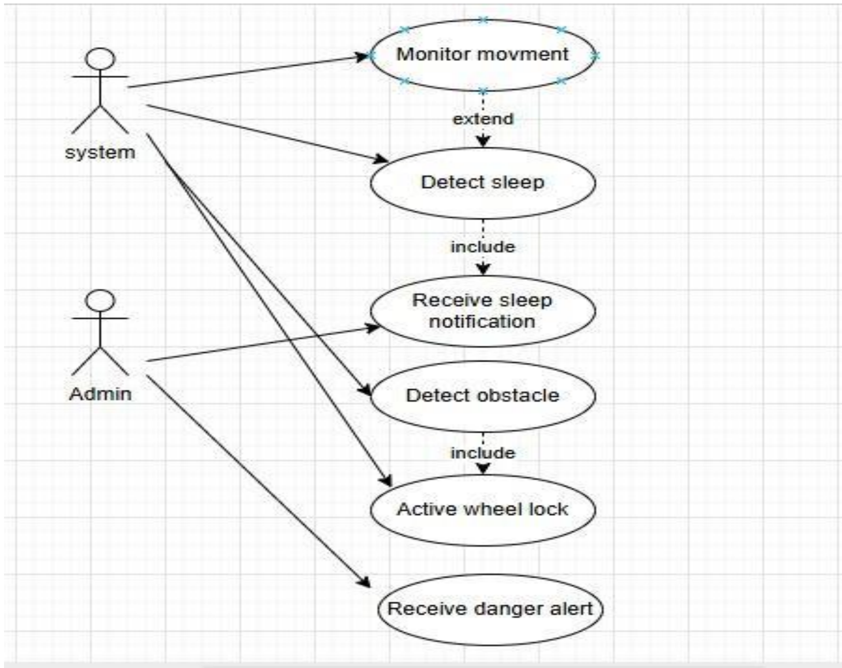


Figure 3-28 Safety monitoring use case

3.4.3 USE CASE TABLE:

Use case	Description
Detect Child in Seat	The system detects whether the child is seated on the stroller using the seat sensor
Trigger Buzzer Alert	The system activates a buzzer sound when the child is detected sitting on the stroller.
Verify Child Identity	The system verifies the child's identity using an NFC tag.
Store Child Data	The system receives and stores the child's information obtained from the NFC tag.
Update Child Record	The system updates the child's stored information when needed.
Sign Up	Allows a user to create a new account.
Login	Allows a user to access the application using their credentials.
View Child Status	Allows the user to view the child's current status.
View Live Camera	Allows the user to watch a live video stream of the child.
Monitor Movement	The system monitors the child's movements using the MPU sensor.
Detect Sleep	The system determines whether the child is asleep or awake based on movement data.
Receive Sleep Notification	The admin receives a notification when the system detects that the child is sleeping.

Detect Obstacle	The system detects obstacles in front of the stroller using an ultrasonic sensor.
Activate Wheel Lock	The system automatically locks the wheels when a dangerous obstacle is detected.

Table 3-1 shows brief description for each use case in our project.

3.4.4 SEQUENCE DIAGRAM:

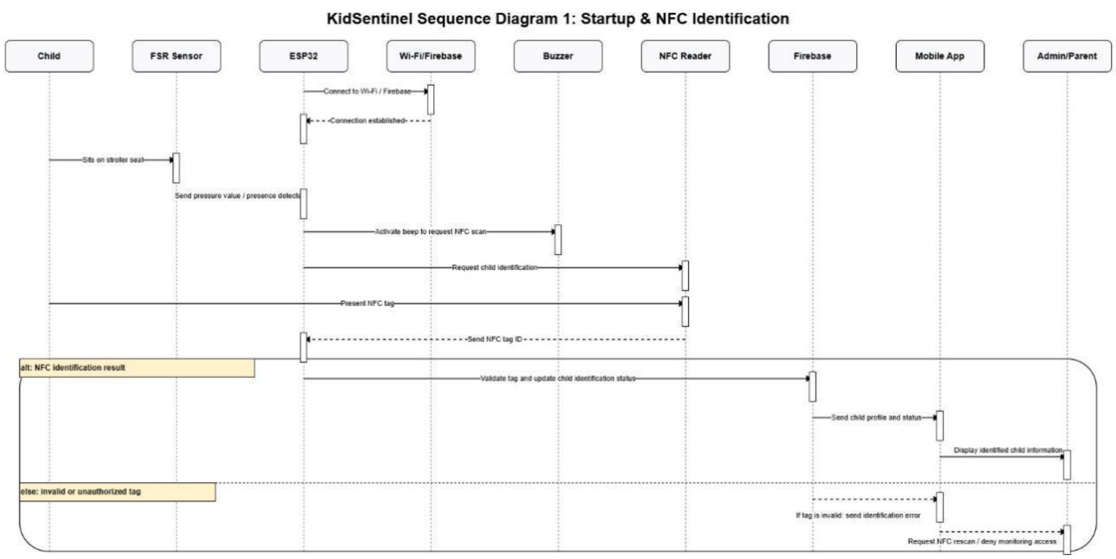


Figure 3-29 Sequence Diagram 1: Startup & NFC Identification

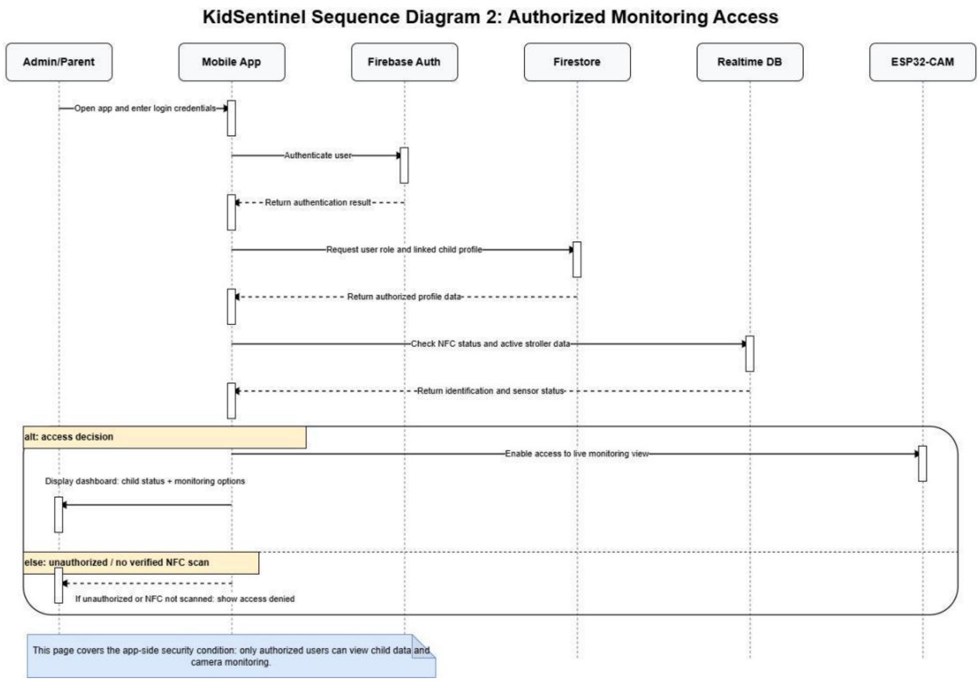


Figure 3-30 Sequence Diagram 3: Live Camera Monitoring

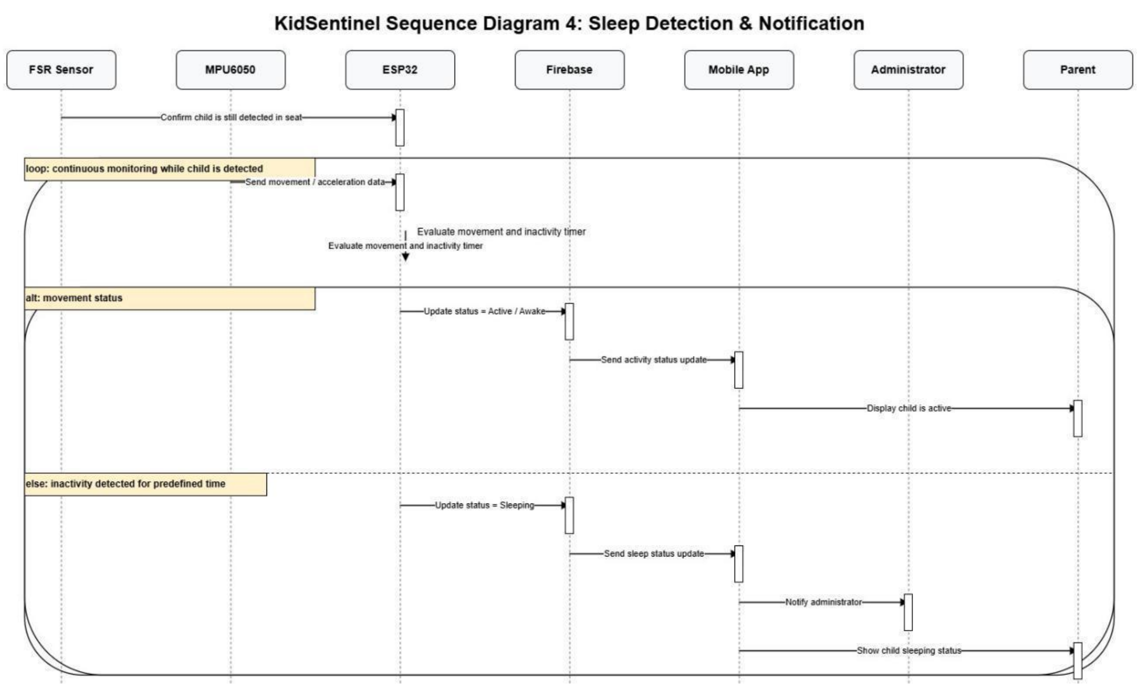


Figure 3-31 Sequence Diagram 4: Sleep Detection & Notification

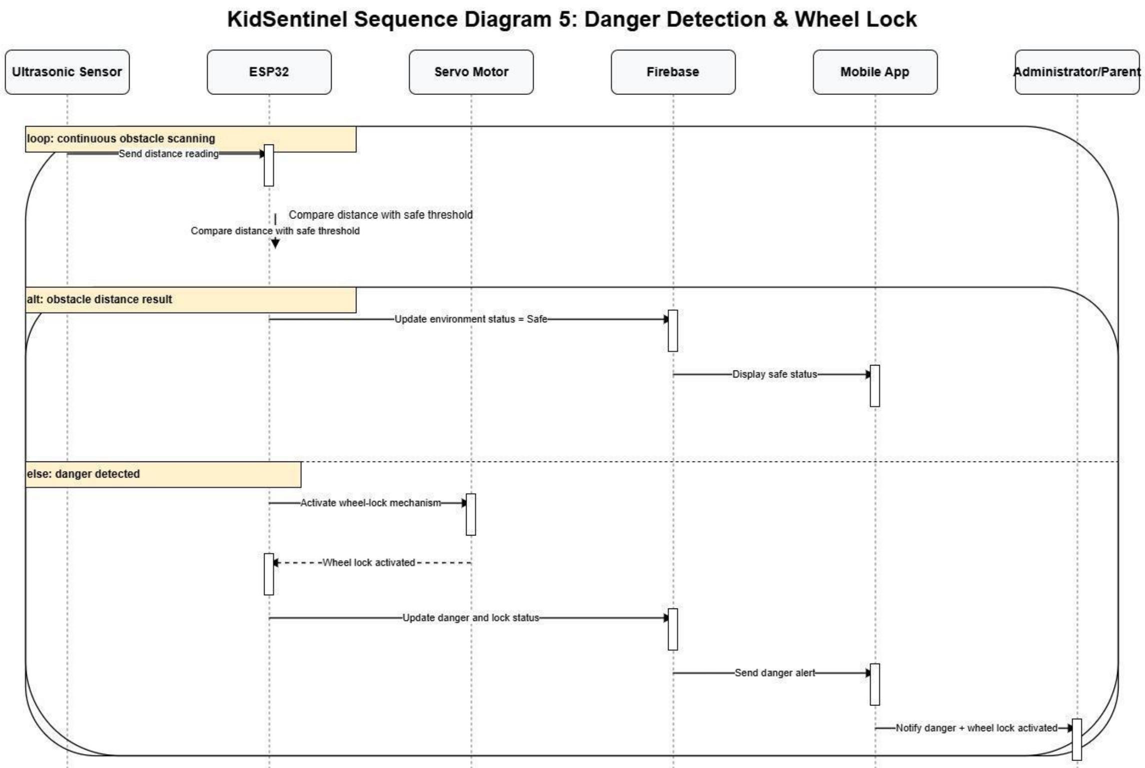


Figure 3-32 Sequence Diagram 5: Danger Detection & Wheel Lock

3.5 Simulation and/or Experimental Test

3.5.1 TEST LOGIN VALIDATION AND VERIFICATION

The system checks your credentials as soon as you log in with your email and password. If the email doesn't exist or the password is incorrect, a warning message appears indicating an error with your login details.

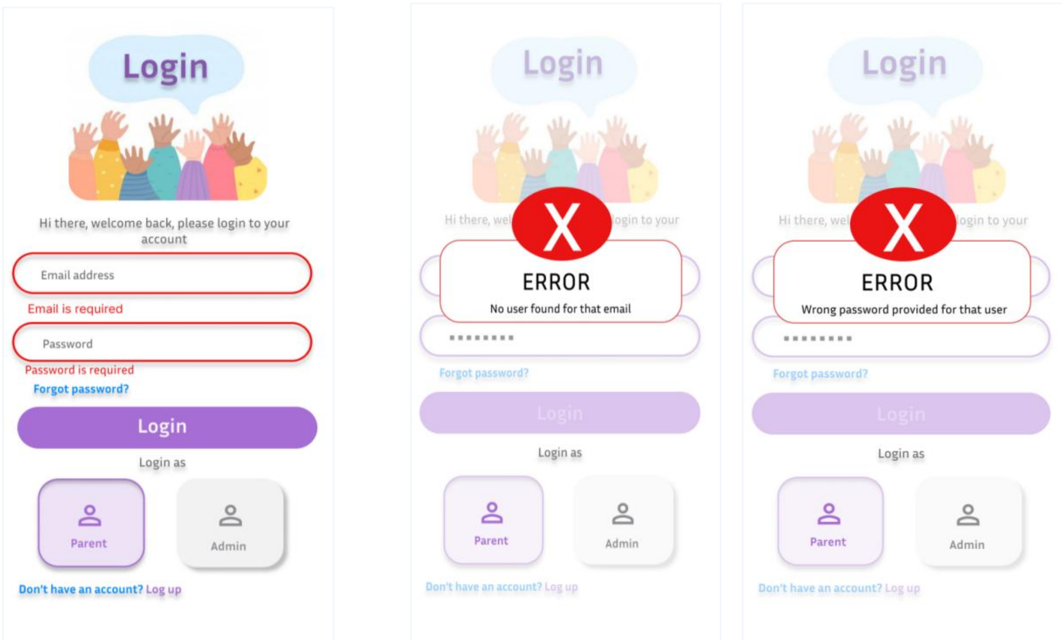
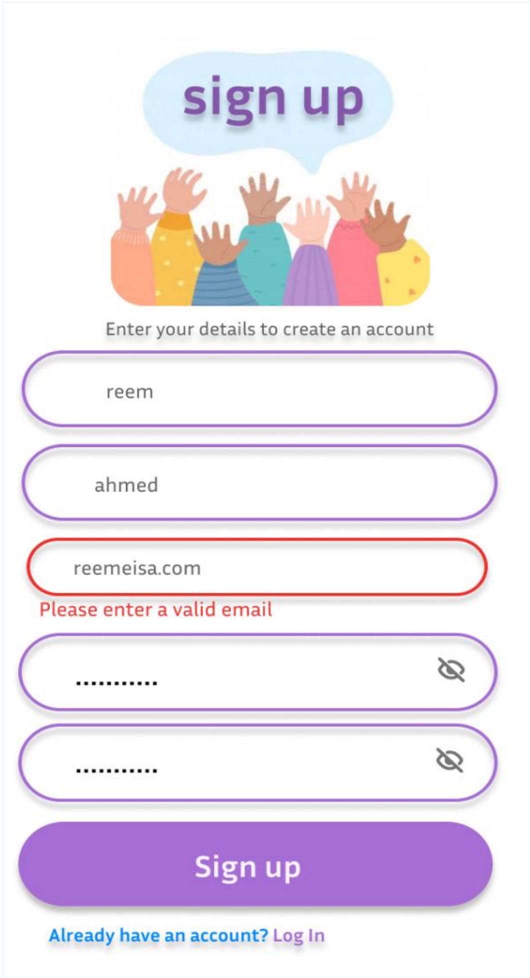
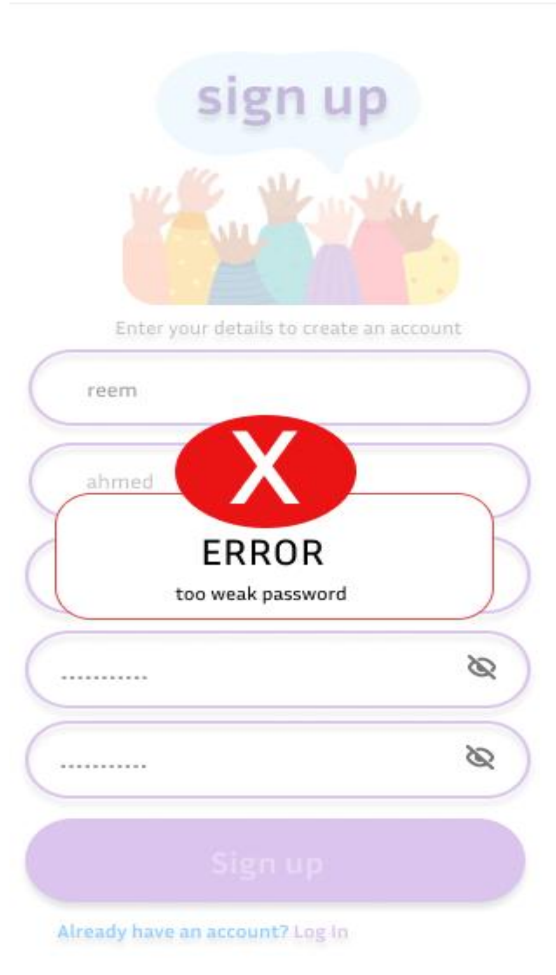


Figure 3-33 Login Verification

3.5.2 TEST SIGNUP VALIDATION AND VERIFICATION

In sign up page, the verification procedure verifies whether there exists a weak password or if the email entered is registered in Firebase. In any of these scenarios, the user is warned about the fact that the email entered is registered and thus cannot be utilized to create an account. Besides, access is only gained after verifying the email entered by the user.


sign up
Enter your details to create an account
reem
ahmed
reemeisa.com
Please enter a valid email
Sign up
Already have an account? Log In


sign up
Enter your details to create an account
reem
ahmed
ERROR
too weak password
Sign up
Already have an account? Log In

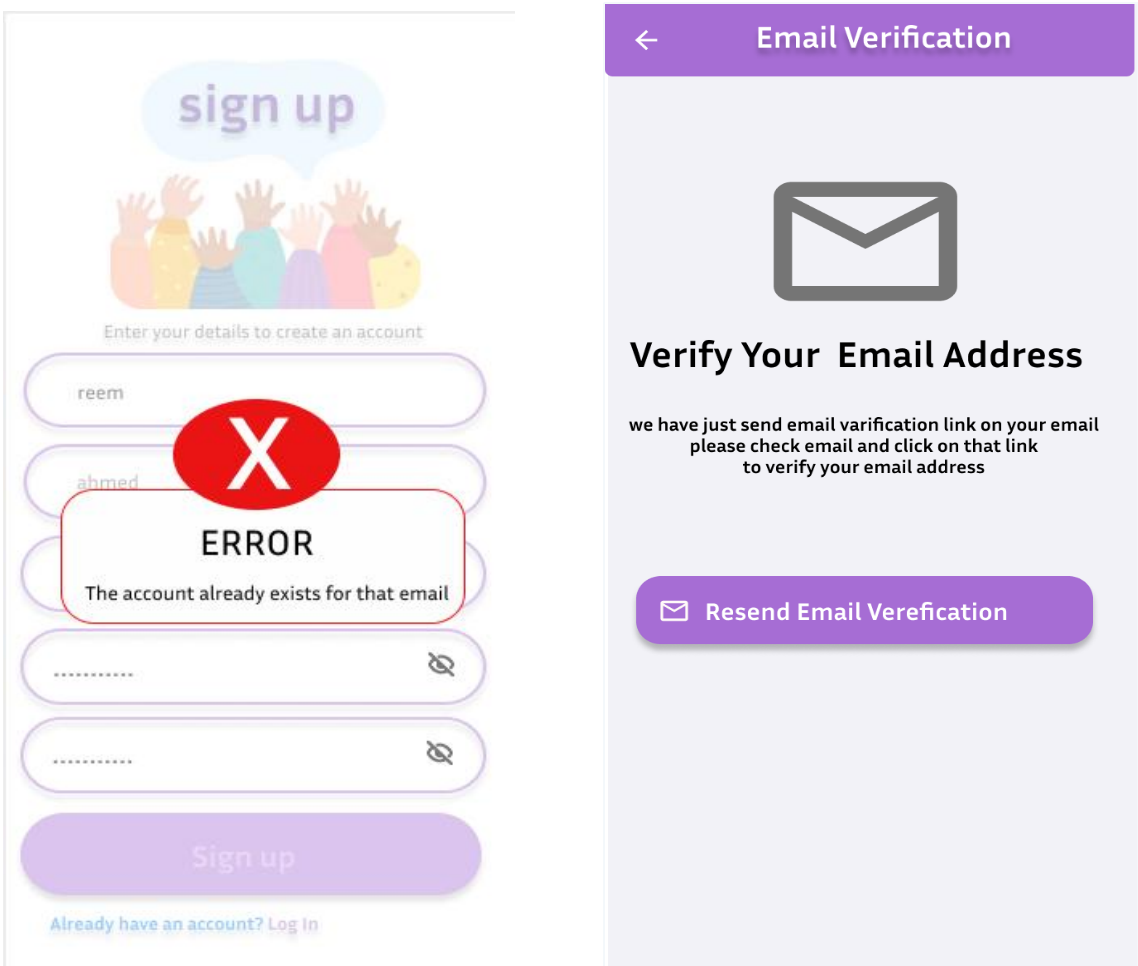


Figure 3-34 Sign up Verification

4 CHAPTER 4: RESULTS AND DISCUSSIONS

4.1 Results

4.2 Discussions

5 CHAPTER 5: PROJECT MANAGEMENT

5.1 Tasks, Schedule and Milestones

The "Tasks, Schedule, and Milestones" section outlines all project activities, their durations, and the team members responsible for each task. It serves as a roadmap that breaks the KidSentinel project down into manageable phases with clear start and end dates, helping the team stay organized and on track throughout the semester. The project officially commenced on March 7, 2026, and is scheduled for completion on June 25, 2026, covering the full Spring 2025/2026 semester.

The project is organized into five main phases: Initiating, Planning, Executing, Monitoring and Controlling, and Closing. The Initiating and Planning phases were completed early in the semester, establishing the project scope, objectives, and resource plan. The Executing phase covers all system design and development activities, including hardware simulation, database design, application construction, and testing. Monitoring and Controlling runs throughout the entire project duration to ensure quality and progress. The Closing phase involves preparing and presenting the final report.

Task Name	Duration	Start	Finish	Predecessors	Resource Names
SENIOR PROJECT I	**81 days**	**Sun Mar 08, '26**	**Wed Jun 24, '26**		**Dana Eisa,Leen Amawi,Rawia Najy**
Initiating	**10 days**	**Sun Mar 08, '26**	**Thu Mar 19, '26**		**Dana Eisa,Leen Amawi,Rawia Najy**
Project kick-off meeting	1 day	Sun Mar 08, '26	Sun Mar 08, '26		Dana Eisa,Leen Amawi,Rawia Najy
Define project idea & scope	5 days	Sun Mar 08, '26	Sun Mar 15, '26	3	Dana Eisa,Leen Amawi,Rawia Najy
Develop project charter	4 days	Sun Mar 15, '26	Thu Mar 19, '26	4	Dana Eisa,Leen Amawi,Rawia Najy
Planning	**15 days**	**Thu Mar 19, '26**	**Thu Apr 09, '26**	**5**	**Dana Eisa,Leen Amawi,Rawia Najy**
Develop project plan	3 days	Thu Mar 19, '26	Tue Mar 24, '26	5	Dana Eisa,Leen Amawi,Rawia Najy
Review & approve project plan	2 days	Tue Mar 24, '26	Thu Mar 26, '26	7	Dana Eisa,Leen Amawi,Rawia Najy
Create scope statement	5 days	Thu Mar 26, '26	Thu Apr 02, '26	8	Dana Eisa,Leen Amawi,Rawia Najy
Define requirements	3 days	Thu Apr 02, '26	Tue Apr 07, '26	9	Dana Eisa,Leen Amawi,Rawia Najy
Resources & cost management	2 days	Tue Apr 07, '26	Thu Apr 09, '26	10	Dana Eisa,Leen Amawi,Rawia Najy
Executing - Senior I	**46 days**	**Thu Apr 09, '26**	**Thu Jun 11, '26**	**11**	**Dana Eisa,Leen Amawi,Rawia Najy**
Related work survey	14 days	Thu Apr 09, '26	Tue Apr 28, '26	11	Dana Eisa,Leen Amawi,Rawia Najy
System overview & design	7 days	Wed Apr 29, '26	Thu May 07, '26	13	Dana Eisa,Leen Amawi,Rawia Najy
Hardware simulation (Fritzing)	7 days	Thu May 07, '26	Mon May 18, '26	14	Dana Eisa,Rawia Najy

Database design (Firebase)	5 days	Thu May 07, '26	Thu May 14, '26	14	Leen Amawi,Rawia Najy
Application UI/UX design	10 days	Thu May 07, '26	Thu May 21, '26	14	Dana Eisa,Leen Amawi
App construction (Sr. I)	10 days	Thu May 21, '26	Thu Jun 04, '26	17	Dana Eisa,Leen Amawi
Initial testing & verification	5 days	Thu Jun 04, '26	Thu Jun 11, '26	15,16,18	Dana Eisa,Leen Amawi,Rawia Najy
Monitoring & Controlling Sr.I	**81 days**	**Sun Mar 08, '26**	**Wed Jun 24, '26**		**Dana Eisa,Leen Amawi,Rawia Najy**
Status reports	57 days	Sun Mar 08, '26	Sun May 24, '26		Dana Eisa,Leen Amawi,Rawia Najy
Performance tracking	57 days	Sun Mar 08, '26	Sun May 24, '26		Dana Eisa,Leen Amawi,Rawia Najy
Closing - Senior I	**10 days**	**Thu Jun 11, '26**	**Wed Jun 24, '26**	**19**	**Dana Eisa,Leen Amawi,Rawia Najy**
Prepare Senior I final report	10 days	Thu Jun 11, '26	Wed Jun 24, '26	19	Dana Eisa,Leen Amawi,Rawia Najy
Submit Senior I report	0 days	Wed Jun 24, '26	Wed Jun 24, '26	24	Dana Eisa,Leen Amawi,Rawia Najy
SUMMER BREAK	**60.5 days**	**Thu Jun 25, '26**	**Tue Sep 15, '26**		**Dana Eisa,Leen Amawi,Rawia Najy**
No project activities	60.5 days	Thu Jun 25, '26	Tue Sep 15, '26	25	Dana Eisa,Leen Amawi,Rawia Najy
SENIOR PROJECT II	**105 days**	**Wed Sep 16, '26**	**Sun Feb 07, '27**		**Dana Eisa,Leen Amawi,Rawia Najy**
Executing - Senior II	**70 days**	**Wed Sep 16, '26**	**Mon Dec 21, '26**		**Dana Eisa,Leen Amawi,Rawia Najy**
Hardware assembly & wiring	14 days	Wed Sep 16, '26	Mon Oct 05, '26	27	Dana Eisa,Rawia Najy
Firmware development (ESP32)	21 days	Mon Oct 05, '26	Tue Nov 03, '26	30	Dana Eisa,Rawia Najy
Mobile app completion	21 days	Wed Sep 16, '26	Wed Oct 14, '26	18	Dana Eisa,Leen Amawi
Firebase integration & testing	14 days	Tue Nov 03, '26	Sun Nov 22, '26	31,32	Leen Amawi,Rawia Najy
Full system integration	14 days	Sun Nov 22, '26	Thu Dec 10, '26	33	Dana Eisa,Leen Amawi,Rawia Najy
Performance evaluation	7 days	Thu Dec 10, '26	Mon Dec 21, '26	34	Dana Eisa,Leen Amawi,Rawia Najy
Monitoring & Controlling Sr.II	**98 days**	**Wed Sep 16, '26**	**Wed Jan 27, '27**		**Dana Eisa,Leen Amawi,Rawia Najy**
Status reports	98 days	Wed Sep 16, '26	Wed Jan 27, '27		Dana Eisa,Leen Amawi,Rawia Najy
Performance tracking	98 days	Wed Sep 16, '26	Wed Jan 27, '27		Dana Eisa,Leen Amawi,Rawia Najy
Closing - Senior II	**35 days**	**Mon Dec 21, '26**	**Sun Feb 07, '27**	**35**	**Dana Eisa,Leen Amawi,Rawia Najy**
Prepare final project report	30 days	Mon Dec 21, '26	Sun Jan 31, '27	35	Dana Eisa,Leen Amawi,Rawia Najy
Final presentation & demo	5 days	Sun Jan 31, '27	Sun Feb 07, '27	40	Dana Eisa,Leen Amawi,Rawia Najy
Project completed	0 days	Sun Feb 07, '27	Sun Feb 07, '27	41	Dana Eisa,Leen Amawi,Rawia Najy

Table 5-1 KidSentinel Project — Tasks, Schedule and Milestones (Task List View)

Table 5.1 KidSentinel Project — Tasks, Schedule and Milestones (Task List View)

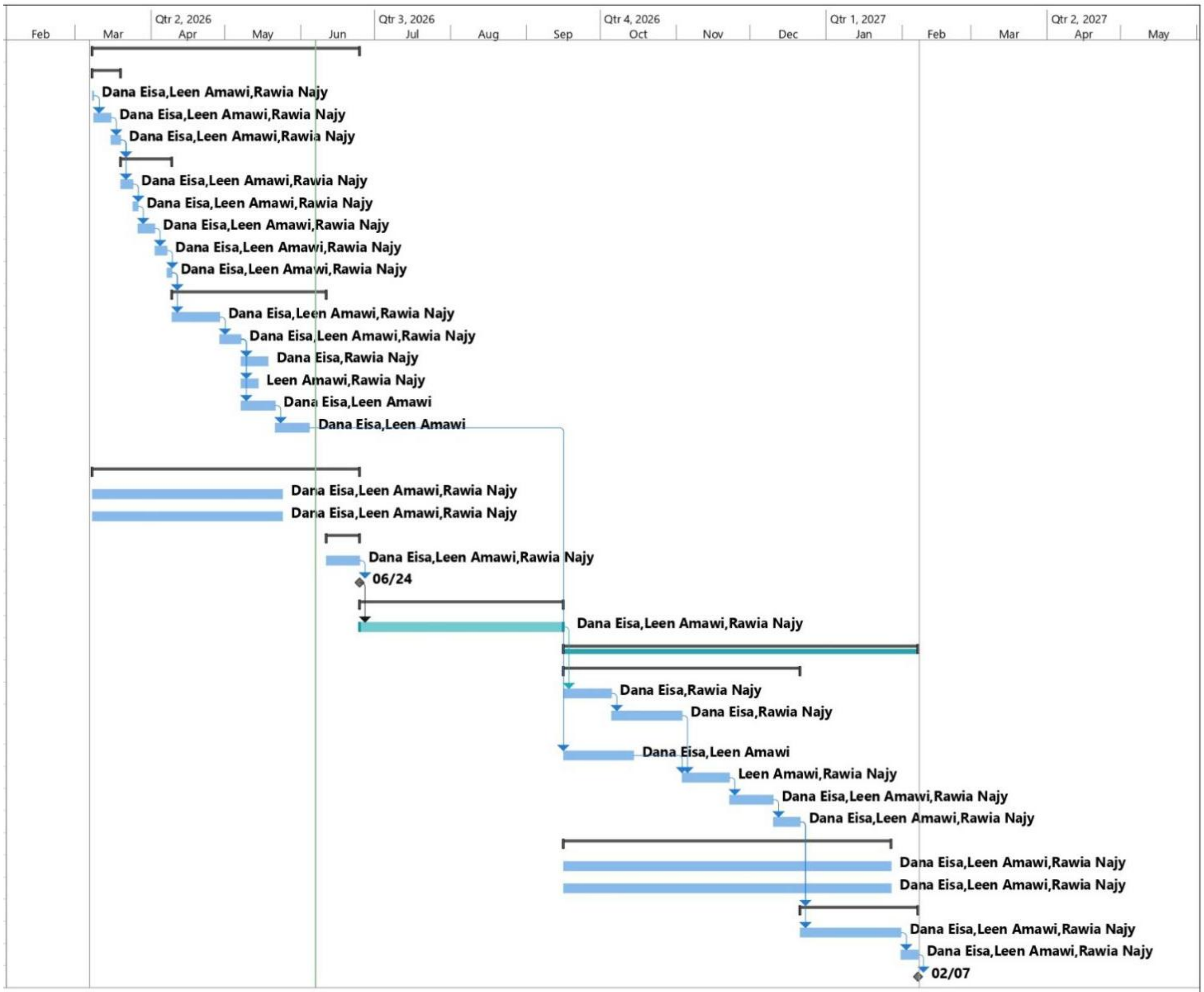


Figure 5-1 KidSentinel Project — Tasks, Schedule and Milestones (Gantt Chart View)

5.2 Resources and Cost Management

Table 5.1 presents the hardware components required to implement the KidSentinel prototype, along with their electrical specifications, unit prices, and total cost in Israeli New Shekels (₪). All prices were collected from local Palestinian electronics suppliers. The total estimated cost for the hardware prototype is ₪392.

Component	Voltage (V)	Current (mA)	Price (₪)	Qty	Subtotal (₪)
ESP32 WROOM-32D	3.3	240	55	1	55
ESP32-CAM Module	5	310	55	1	55
MPU-6050 Sensor	3.3	3.9	32	1	32
FSR Sensor	3.3–5	—	50	1	50
Ultrasonic HC-SR04	5	15	20	1	20
NFC RC522 Kit	3.3	13	40	1	40
Servo Motor SG90	4.8–6	100	55	1	55
Passive Buzzer	3–5	30	5	1	5
Jumper Wires (set)	—	—	20	1	20
Power Bank (5V/2A)	5	2000	60	1	60
				Total Cost	392

Table 5-2 Resources and Cost Management

5.3 Lessons Learned

Throughout the development of the KidSentinel project, the team gained valuable insights spanning both technical and professional domains. On the hardware side, working with the ESP32 microcontroller deepened the team's understanding of embedded systems, including how to configure GPIO pins, manage I2C communication for the MPU-6050 motion sensor, and interface with the NFC reader. Practical experience was also gained in sensor integration, including calibrating the FSR pressure sensor and the ultrasonic sensor, and controlling actuators such as the servo motor using PWM signals.

On the software side, the team expanded its skills in mobile application development using Dart and the Flutter framework. Significant experience was gained in designing multi-role interfaces that serve both parents and nursery administrators, as well as in connecting the application to Firebase Realtime Database and Firestore for live data exchange. The team also learned how to structure cloud data effectively, separating sensor status data from user and child profile data across different Firebase collections.
Beyond the technical aspects, the project reinforced the importance of effective teamwork, clear communication, and structured project planning. Dividing responsibilities among team members according to their strengths — such as separating hardware integration tasks from application development — proved essential for maintaining steady progress. Pre-project planning, literature review, and studying related works helped the team avoid common

pitfalls and make informed design decisions. Overall, this project provided a comprehensive and practical engineering experience that combines IoT hardware, cloud computing, and mobile software development in a real-world child safety context.

6 CHAPTER 6: IMPACT OF THE ENGINEERING SOLUTION

6.1 Economical, Societal and Global

Since there are smaller electronic components involved in this system, it is relatively economical when compared to other systems due to the lower amount of energy needed and the fewer resources that will be consumed in its operation.

Socially, the system ensures increased security for children through offering a continuous monitoring process and alerting the administrator about unusual movements of the kids. Moreover, the system offers more confidence for parents about the well-being of their children. Besides, the system raises awareness about using smart systems like IoT applications in order to keep the child safe.

Globally, the proposed project fits in the ongoing trend related to the application of IoT technology in all life fields. Moreover, taking advantage of any new development of technology in order to provide better solutions for enhancing children's safety globally.

6.2 Environmental and Ethical

The system minimizes its impact on the environment by using energy-efficient parts, leading to a reduction in the amount of energy used and thus helping towards building an environmentally-friendly IoT system.

Ethically, the system guarantees children’s privacy as only authorized individuals will have access to it, and no form of data leakages will take place. Furthermore, data gathered from the system will only be used for monitoring purposes.

7 CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Summary of Achievements of the Project Objectives

The main goal of this project was to create a child monitoring system that's smart. This child monitoring system uses both hardware and software to make sure children are safe in nurseries. The system is made to help the people who work at the nursery watch the children better. They can use a phone app that is connected to the hardware with Wi-Fi.

The system we made uses a tag to know who each child is. It also uses sensors to check on children. One sensor checks if the child is sitting in the stroller. Another sensor checks if the child is moving around. If the child is sleeping, the system sends a message to the nursery staff on their phone.

To keep the children safe, the system uses a sensor to find things in the way of the stroller. If something is too close, the system stops the stroller to prevent accidents.

The system also has a camera on the stroller. This camera lets the nursery staff see the child on their phone. We used something called Firebase to help the phone app talk, to the hardware and store information.

The project did what it was supposed to do. It made a system that can identify children check if they are sitting down and see if they are asleep, find things in the way stop the stroller show video of the child and send messages to the nursery staff. This system is smart and easy to use. The child monitoring system is a way to keep children safe in nurseries.

7.2 New Skills and Experiences Learnt

We learned a lot of things while working on this project. We found out about monitoring systems and how they can be used in childcare. This was really useful because we got to see how new technology can help keep kids safe.

We learned about the parts that make up these systems like ESP32 microcontrollers, NFC modules, seat sensors, MPU motion sensors, ultrasonic sensors, and camera modules. We also figured out how to connect all these parts to make a complete system that can monitor kids.

We looked into the software side of things too, like using Flutter to make mobile apps and Firebase to manage data and help the different parts of the system talk to each other. This helped us understand how all the different technologies work together in the system we were proposing.

Working on the project also taught us skills like doing research, writing about technical things, gathering information and analyzing systems. We got to practice planning a project deciding what we needed to do and designing the system before we started building it. The project documentation was really helpful for learning about monitoring systems and all the new skills we gained from working on it like new skills and experiences, with smart monitoring systems.

7.3 Recommendations for Future Work

We hope that the child monitoring system we made can be used in nurseries one day. The child monitoring system needs work to make it better and safer for kids.

We think it would be an idea to add more features to the child monitoring system. One thing we can do is make the child monitoring system smarter by using intelligence to watch what the kids are doing and send alerts when something is wrong. The child monitoring system can also keep track of things that happen and make reports to help the people who work at the nursery take care of the kids.

The part of the child monitoring system that lets people watch the kids in time can be improved too. We can make the video better. Let people see what is going on with multiple cameras using the app. The child monitoring system can also be made to work with kids and strollers in the same nursery.

We think that using technology in places where kids are taken care of can make things safer and more efficient. This can give the people who work at the nursery and the parents of the kid's peace of mind. The child monitoring system can be a good tool for the nursery staff and parents if we keep working on it.

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APPENDICES

9 APPENDIX A:

10 APPENDIX B:

11 APPENDIX C: